Open-Source Analog IC Design Workshop

1.1 Introduction to open-source integrated circuit design

Jorge Marin N., PhD

Associate Professor, EIE-PUCV

Research Associate, AC3E-UTFSM

UFRO, Temuco – November 10-11, 2026

Con el apoyo de Proyecto ANID Fondecyt de Exploración 2025 + 13250147

ESCUELA DE INGENIERÍA ELÉCTRICA — PONTIFICIA UNIVERSIDAD CATÓLICA DE VALPARAÍSO

UNIVERSIDAD TECNICA FEDERICO SANTA MARIA — AC3E

SEED: SEMICONDUCTOR EDUCATION, EMPOWERMENT, & INDUSTRY DEVELOPMENT

IEEE SSCS Solid-State Circuits Society IC Innovation — Electron Devices Society

Jorge Marin N.



➢ Academic Degrees

- Ph.D. in Electrical Engineering. KU Leuven, Belgium, 2018.
  - KU Leuven Doctoral fellowship + IWT-Flanders research project grant
- MsC degree in Electrical Engineering. KU Leuven, Belgium, 2012.
- BsC/MsC in Electrical Engineering. Universidad de Chile, Santiago, Chile, 2010.

➢ Previous Work Experience

- Associate Professor at PUCV, Valparaíso, Chile, 2025-Today.
- Research Associate at Advanced Center for Electrical and Electronic Engineering (AC3E), USM, Valparaíso., Chile, 2023-Today.
- Postdoctoral Research Fellow at AC3E-USM, Valparaíso, Chile, 2022-2023.
- Analog Designer. Sony Depthsensing Solutions, Belgium, 2020 – 2021.
- Analog Designer. ICSense (TDK), Belgium, 2018 – 2020.
- Engineer Intern/ Engineer. Synopsys R&D Center, Santiago, Chile 2006 - 2009.
- Lecturer and TA, 2008 - 2025
  - Universidad de Chile, Universidad de los Andes (Chile), KU Leuven, USM
  - Digital Systems, Analog Integrated Circuit Design, Analog IC Design Lab

➢ Other roles

- IEEE CASS/SSCS/EDS joint chapter chair
- IEEE CICC TPC member
- UNIC-CASS educational program team lead

Introduction

Introduction

Semiconductor chips are transforming world economy



A microchip is a set of electronic circuits on a small flat piece of semiconductor material, usually silicon.

A microchip the size of a fingernail can contain billions of transistors, which are electronically controlled switches.



→ 1 smartphone contains around **160 chips**

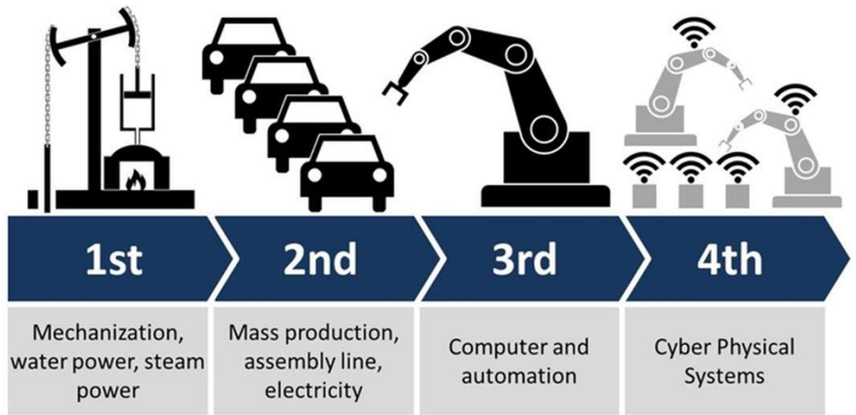
→ 1 hybrid electric car contains up to **3 500 chips**

https://www.consilium.europa.eu/en/infographics/eu-chips-act/

Introduction

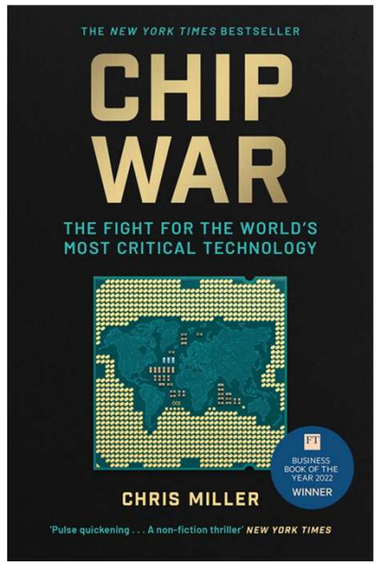
Why chips matter today?

Schwab, The Fourth Industrial Revolution [2017]



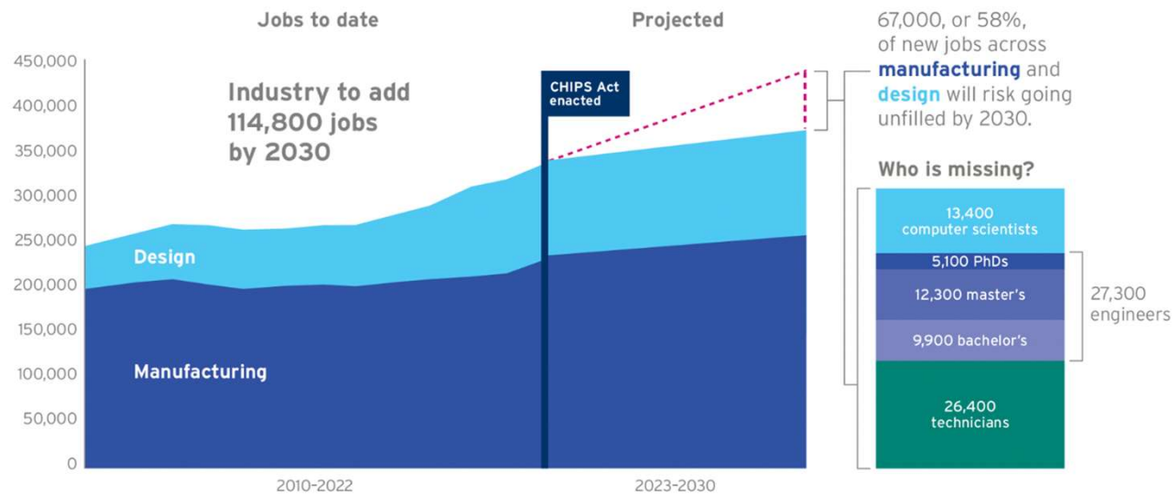
High demand for complex chips → exponential technologies

[2022]



Geopolitical context →chips are the new oil!

SIA, Chipping Away [2023]



Workforce crisis → CS is (way) more interesting than EE

Introduction

What about Chile?



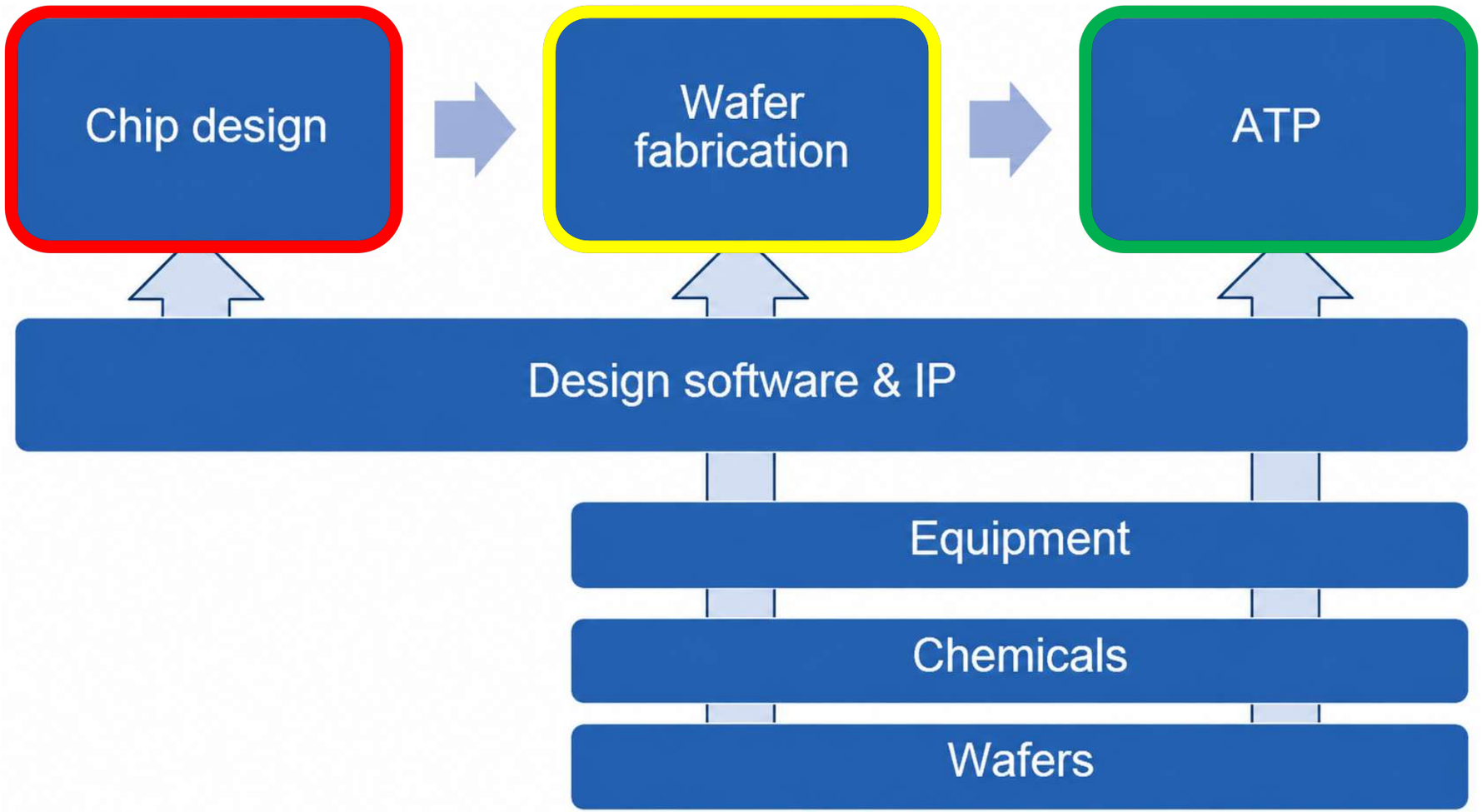
Académico aborda la importancia geopolítica de los semiconductores

Historiador estadounidense Chris Miller explica la guerra de los chips

Fuente: https://www.lun.com/Pages/NewsDetail.aspx?dt=2026-05-06&SupplementId=0&PaginaId=10&EsAviso=0

Introduction

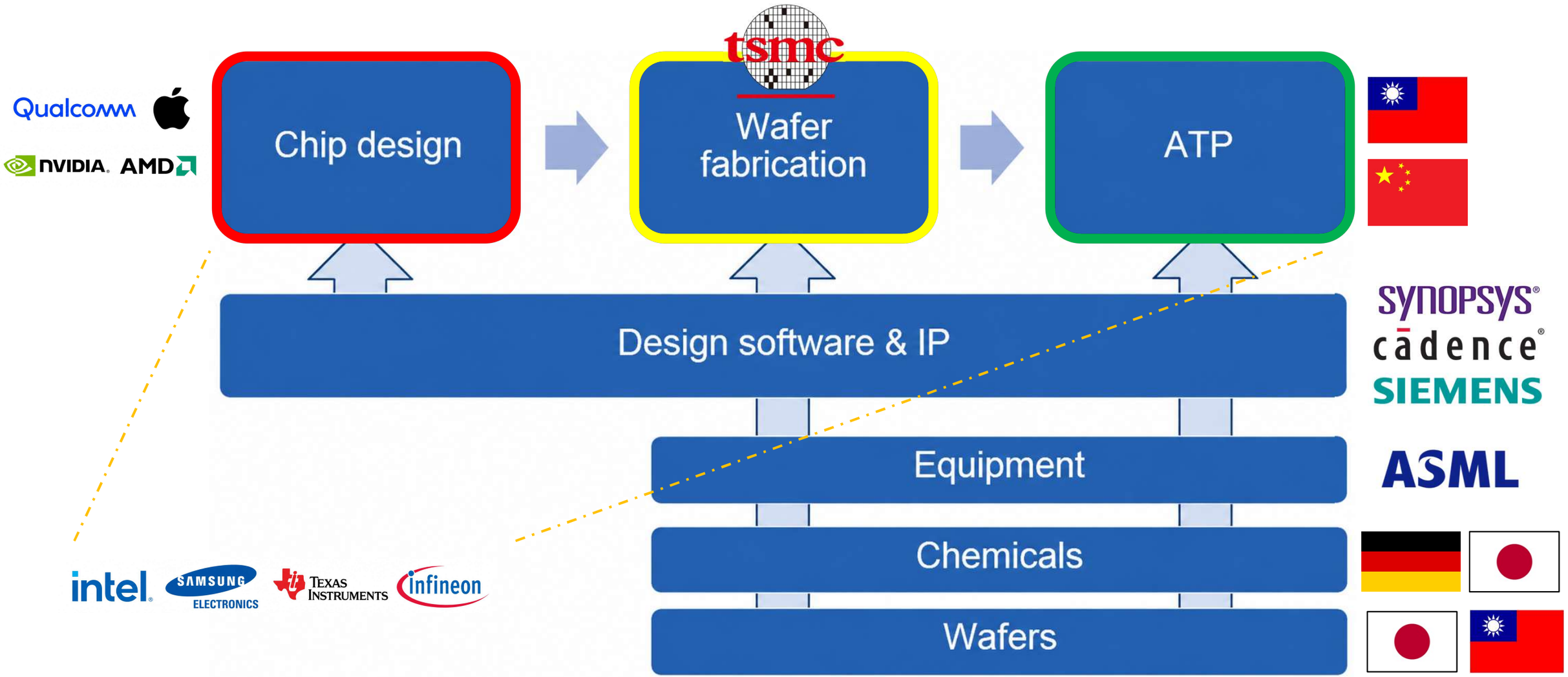
The semiconductor value chain (in a nutshell)



https://www.oecd.org/en/publications/mapping-the-semiconductor-value-chain_4154cdbf-en.html

Introduction

The semiconductor value chain (in a nutshell)



https://www.oecd.org/en/publications/mapping-the-semiconductor-value-chain_4154cdbf-en.html

Introduction

"Harder/better/faster/stronger"

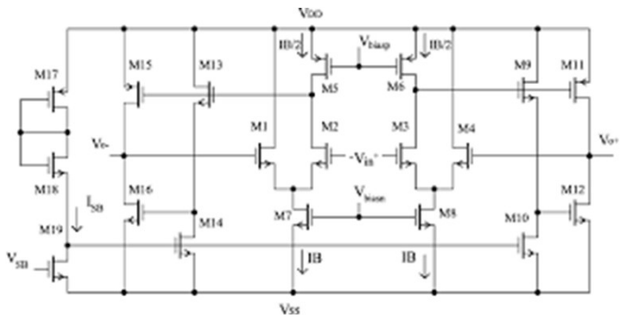
→ We need to

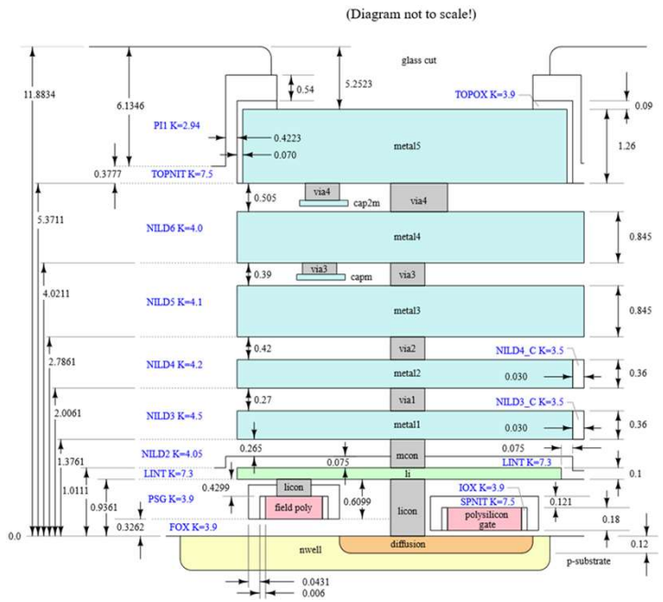
- Design more chips
- Make them more complex, reliable and robust
- Do this with less engineers

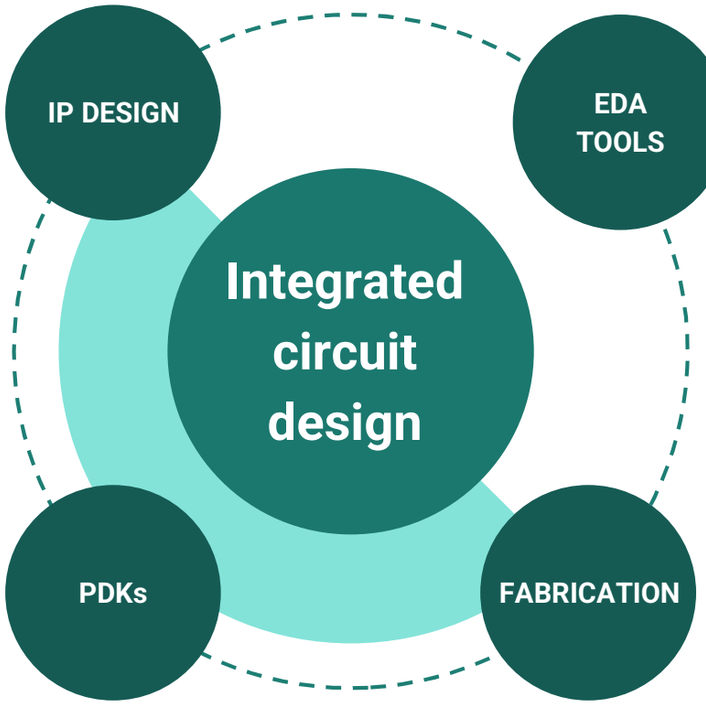
The current IC design ecosystem is not keeping the pace!

Introduction

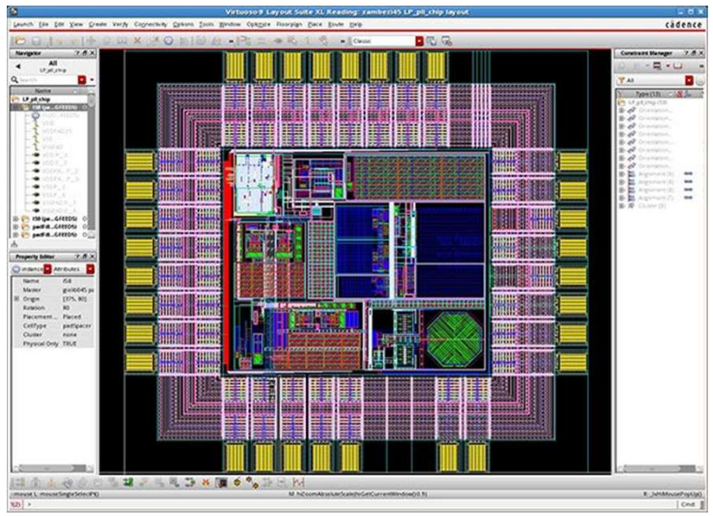
Why Open-Source IC design?

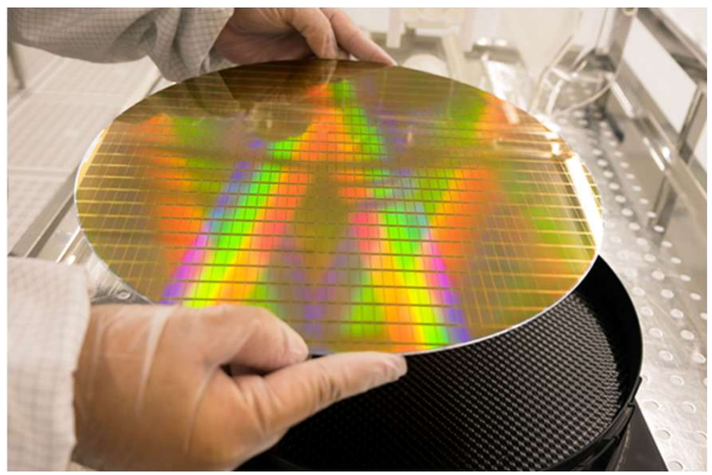






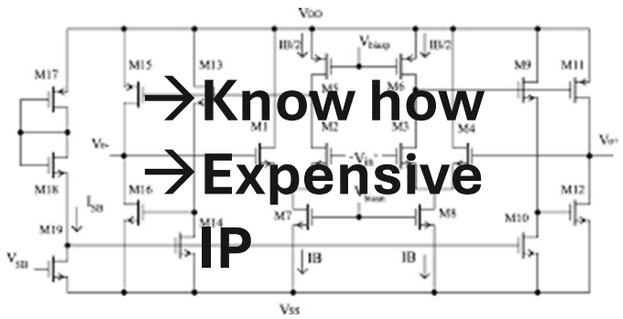
[Tim Ansell, FOSSi Dial-Up2020]



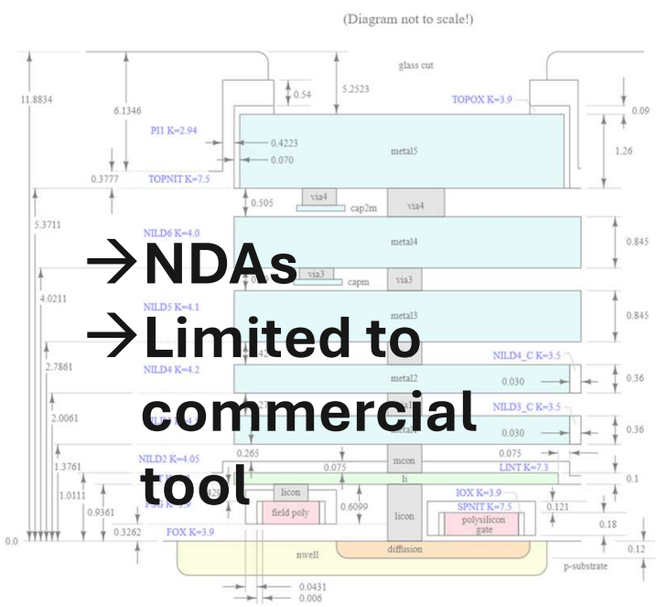


Introduction

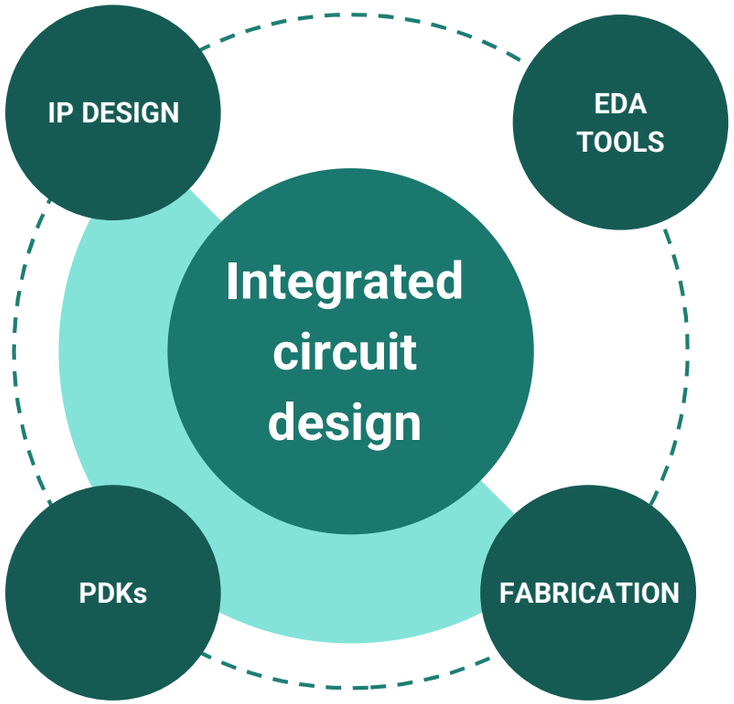
Why Open-Source IC design?



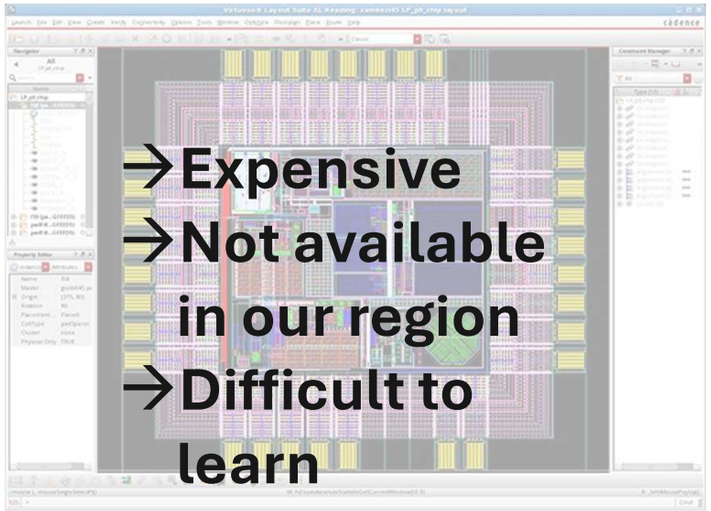
→Know how
→Expensive
IP



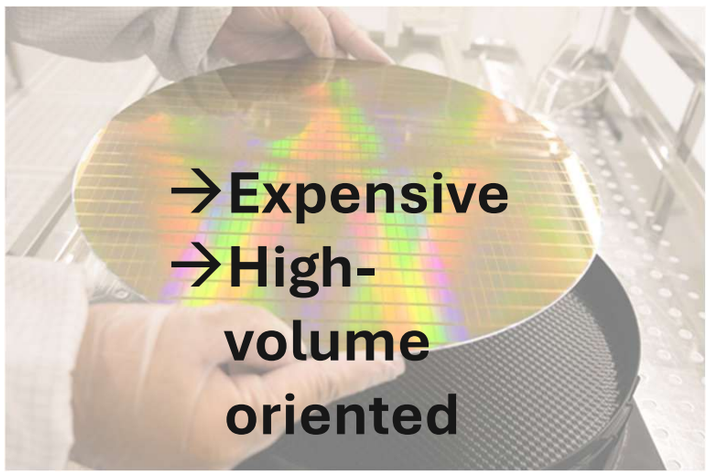
→NDAs
→Limited to commercial tool



[Tim Ansell, FOSSi Dial-Up2020]



→Expensive
→Not available in our region
→Difficult to learn



→Expensive
→High-volume oriented

Introduction

Concern and Chips Acts


The New York Times

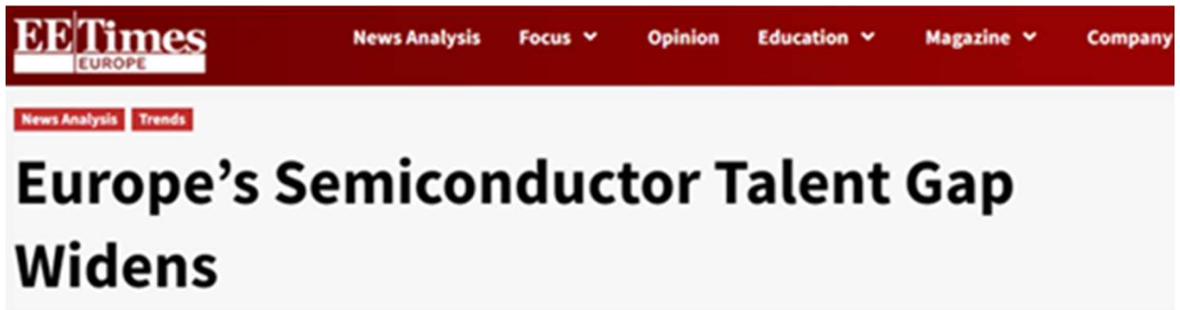
America's Semiconductor Boom Faces a Challenge: Not Enough Workers

Strengthened by billions of federal dollars, semiconductor companies plan to create thousands of jobs. But officials say there might not be enough people to fill them.


Le Monde

« Une partie des candidats qu'on cherche n'existent pas » : une entreprise face aux difficultés de recrutement

Les difficultés d'embauche du fabricant de semi-conducteurs X-Fab révèle combien la question est complexe.


EE Times EUROPE

News Analysis | Focus | Opinion | Education | Magazine | Company

News Analysis Trends

Europe's Semiconductor Talent Gap Widens

Introduction

Concern and Chips Acts

Exploring the 2022 CHIPS and Science Act

Ensuring increased U.S.-based production and improved supply chain resiliency of semiconductor technology

$52.7 Billion in New Funding for Semiconductors



U.S. Chips and Science Act

EU Chips Act

The EU chips act aims to:

- → ensure **large-scale capacity building** and **innovation** within the EU
- → ensure that the EU is **self-supplying** to a much greater extent
- → ensure that the EU can **react quickly** in the event of supply crises



In total, more than €43 billion of policy-driven investment will support the Chips Act until 2030, which will be broadly matched by long-term private investment.

Introduction

Concern and Chips Acts?

Trump renegotiating Biden-era Chips Act grants, Lutnick says

By **Reuters**

June 4, 2025 11:39 PM GMT-4 · Updated June 4, 2025

Reuters

The European Chips Act has no clothes

Opinion | May 3, 2025

By Peter Clarke

EUROPEAN CHIPS ACT | ANALOG | ARTIFICIAL INTELLIGENCE | SEMICONDUCTOR | AUTOMOTIVE

Economy | Technology

Trump tells US chip design software makers to halt China sales: Report

ALJAZEERA

US electronic design automation software makers were told via letters to stop supplies to China, the FT reported.

Introduction

Concern and Chips Acts?

BBC

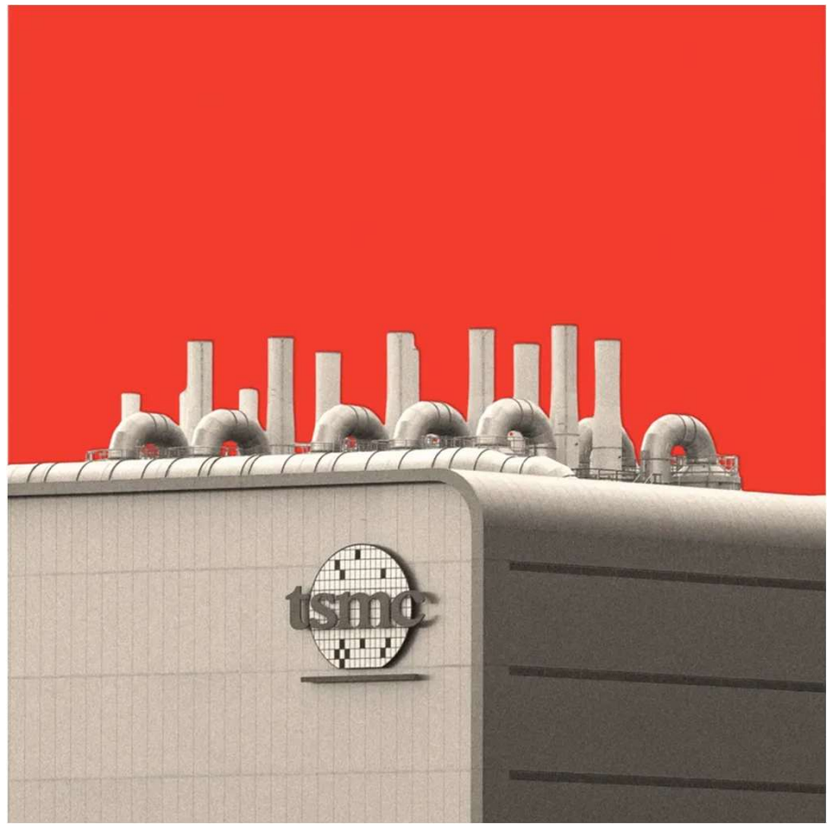
The secretive US factory that lays bare the contradiction in Trump's America First plan

18 May 2025

Share Save

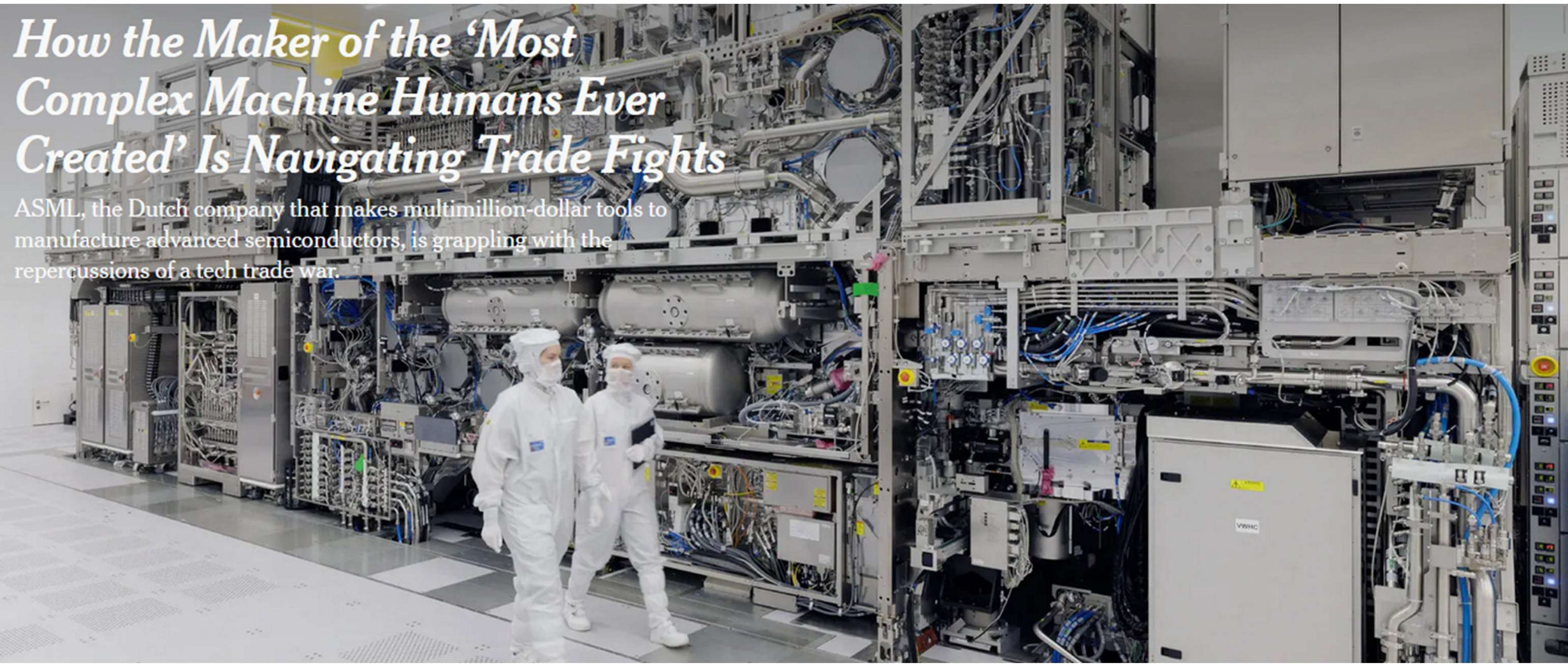
Faisal Islam
Economics editor • @faisalislam

Reporting from Arizona



Introduction

Concern and Chips Acts?



Open-Source Silicon Approach

Open-Source Silicon Approach

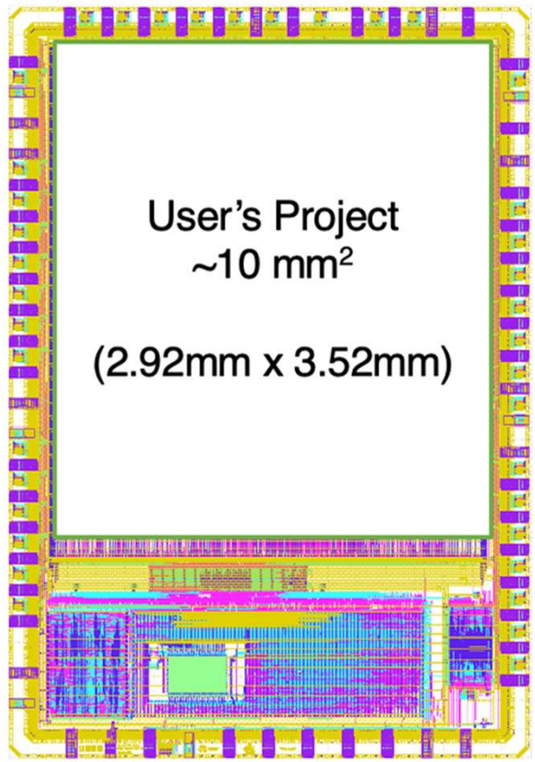
The OS approach: open SKY130 PDK (2020)

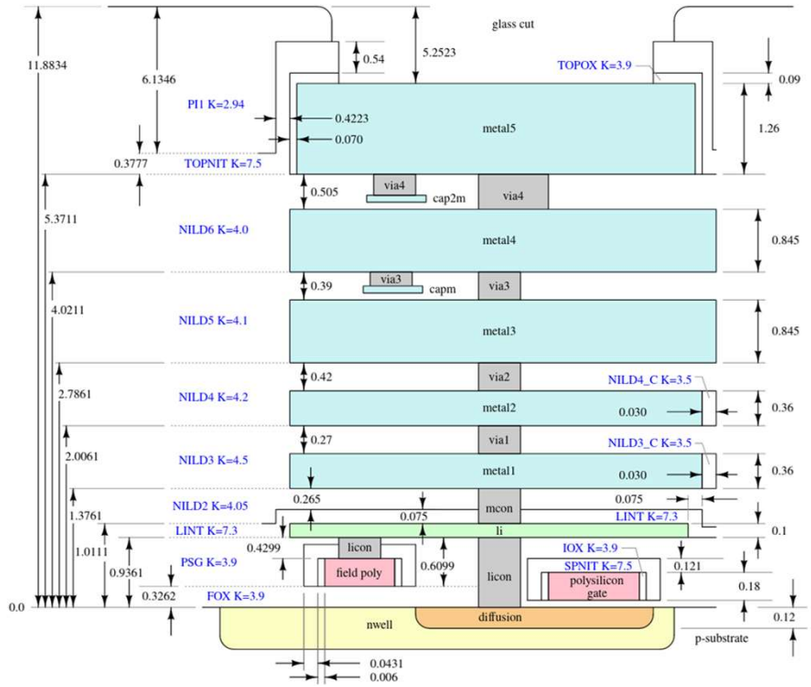
efabless.com

Google + skywater TECHNOLOGY

FOSS 130nm Production PDK
github.com/google/skywater-pdk

Caravel/ Caravan chip harness

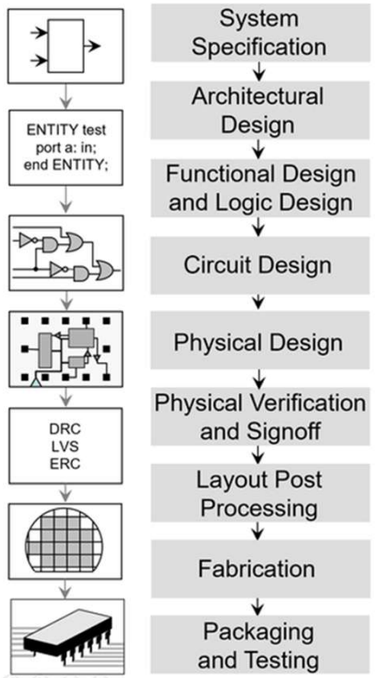




Open-Source Silicon Approach

The OS approach: opensource EDA/CAD software

DIGITAL FLOW



System Specification
↓
Architectural Design
↓
Functional Design and Logic Design
↓
Circuit Design
↓
Physical Design
↓
Physical Verification and Signoff
↓
Layout Post Processing
↓
Fabrication
↓
Packaging and Testing

Yosys OpenROAD 24h

ANALOG FLOW

NGSPICE

Xyce PARALLEL ELECTRONIC SIMULATOR

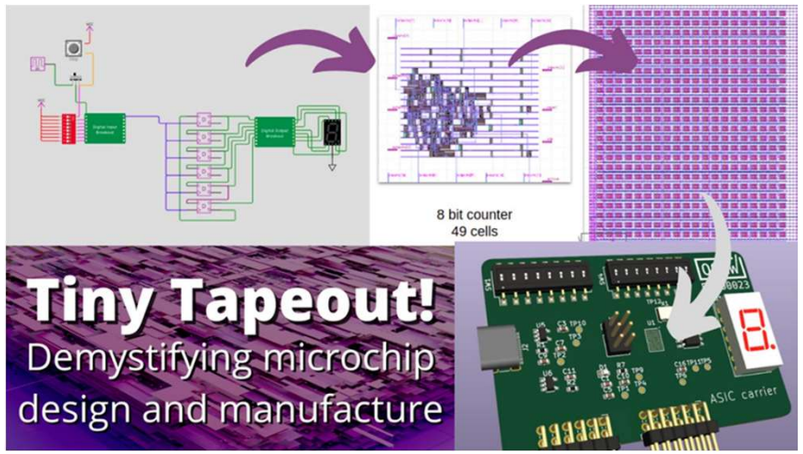
SPECIFICATIONS
↓
SCHEMATIC DESIGN & SIMULATION
↓
LAYOUT GENERATION
↓
DRC & LVS
↓
PARASITIC EXTRACTION & SIMULATION
↓
GDS GENERATION & SUBMISSION

XSCHEM

Magic VLSI Layout Tool
Current distribution version 8.2

Open-Source Silicon Approach

The OS approach: affordable silicon



chipfoundry

ChipCreate

Chip Design, Fabrication and Bring-up for Product Companies, Startups and university programs

$14,950 per tapeout

The ChipCreate offering includes:

- Pre-built SoC design with RISC-V subsystem and peripherals
- Up to 15mm² of die space with a standard I/O ring
- 38 fully -configurable I/Os supporting both digital and analog signaling
- Option of 100 QFN-packaged parts or Bare Die
- Plug and play development board with software support
- Complete RTL-to-GDSII Open Source design flow

ihp Leibniz Institute for High Performance Microelectronics

Open-Silicon MPW Access – Democratizing Chip Design

IHP's open-silicon Multi-Project Wafer (MPW) program provides an affordable path to fabricate your designs using our advanced technologies. Whether you're a startup, researcher, or open-source developer — submit your design and bring your ideas to life in silicon.
Apply now to secure your spot in the next MPW run!

wafer.space

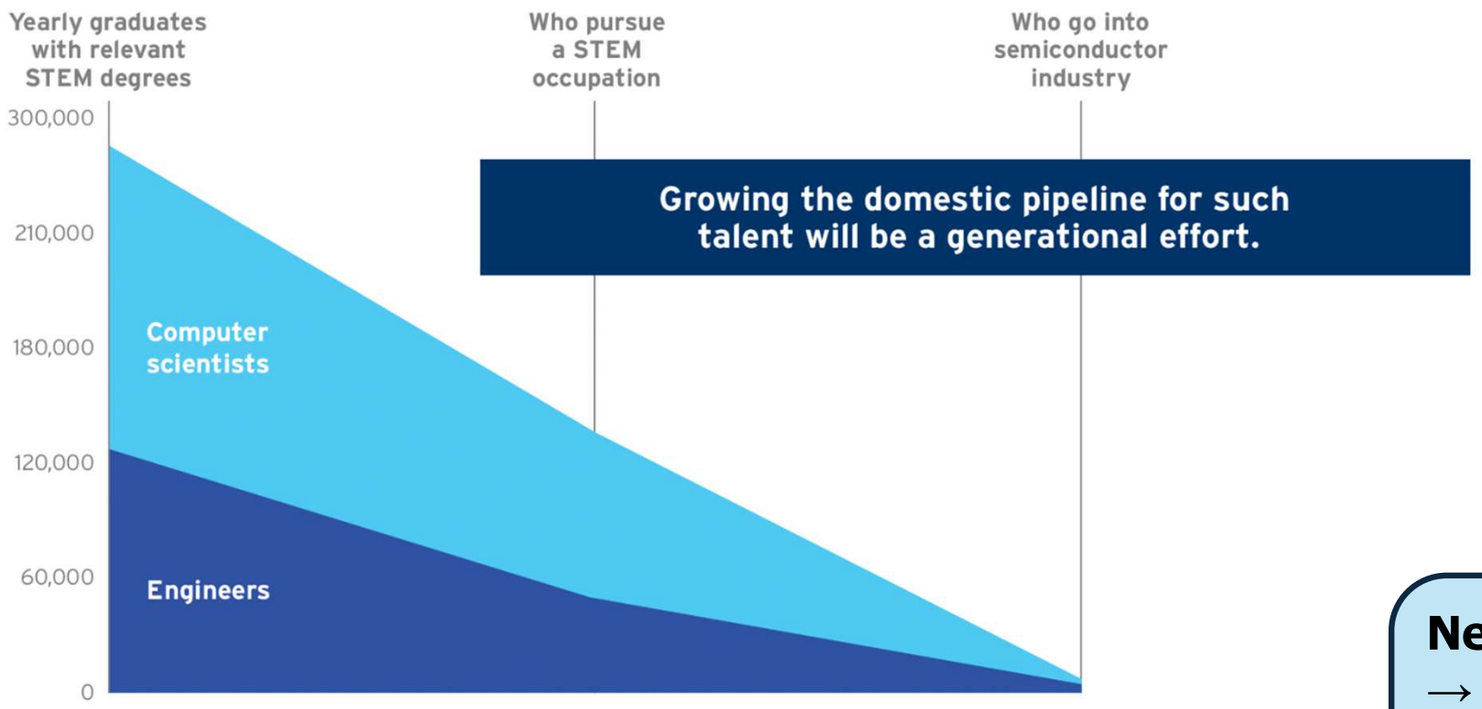
QUARTER SLOTS ARE HERE!

Get 1,000 custom silicon dies from just $2 per die

New quarter slot (0.5×0.5) option — 1,000 dies for $2,000 with Early Bird pricing.

Open-Source Silicon Approach

The OS approach: low barriers for new designers



Source: https://www.semiconductors.org/chipping-away-assessing-and-addressing-the-labor-market-gap-facing-the-u-s-semiconductor-industry/

USA career choice
CS → 830.000
EE → 70.000

Nearterm solution → Attract international talent

New skills in designers:
→ Analog + digital background
→ AI/ML knowledge
→ Programmatic/systematic design skills

Open-Source Silicon Approach

The OS approach: low barriers for new designers





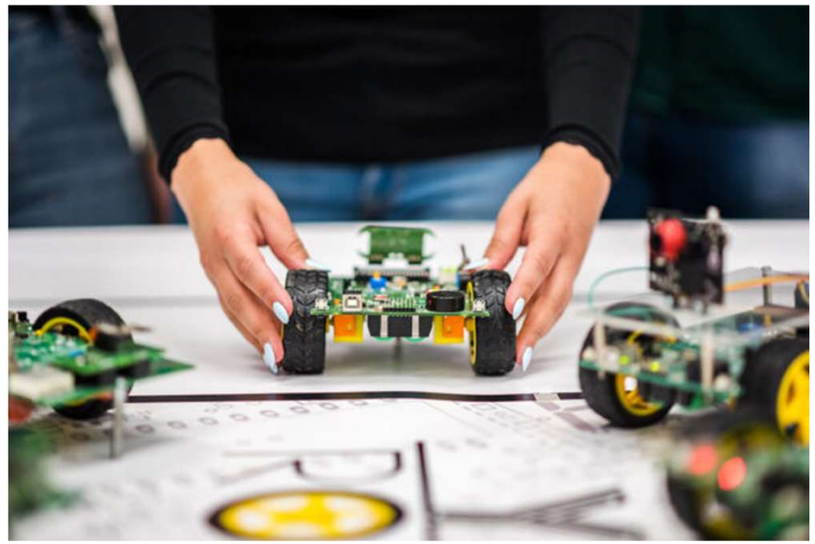
Open-Source Silicon Approach

The OS approach: low barriers for new designers



The IC design workflow is conservative → need for openness and collaboration!

Makers' philosophy: do it yourself



GitHub slack python jupyter

Open-Source Silicon Approach

Industrial initiatives for OS silicon education

CHIPS ALLIANCE

AMD antmicro FUTUREWEI Technologies Google

intel Microsoft SiFive VeriSilicon

Infineon

SUMMERSCHOOL 2023
18-22 SEPTEMBER

AnaGen

Next level
chip design!

IPCEI on Microelectronics



ihp Leibniz Institute for High Performance Microelectronics

Certificate Course: Analog Design with IHP SG13G2 Open-Source PDK

Tutorials | Technologies for Smart Systems 05/19/2025

cadence

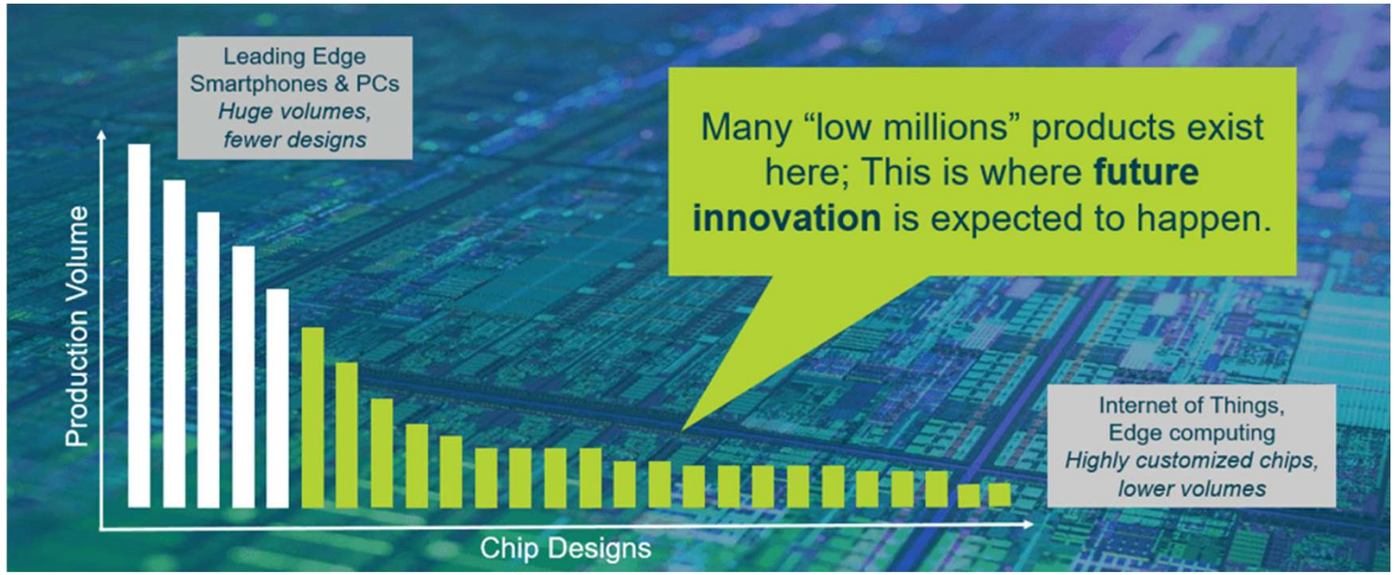
Cadence and SkyWater: Fostering the Next Generation of Innovators

26 May 2025 • 3 minute read

Open-Source Silicon Approach

Industry: why to care about opensource?

- Some pioneer companies are starting to build their business models around open source silicon movement
- Mostly focused on education, OS EDA and enabling chip design for low-volume projects
- But some of them are already working on design (Mabrains, Spherical), and soon more will come



Source: https://www.skywatertechnology.com/sky130-open-source-pdk/

chipfoundry

Tiny Tapeout

Zero to ASIC

SPHERICAL

VLSI System Design

Mabrains

lowRISC

ChipFlow

YosysHQ

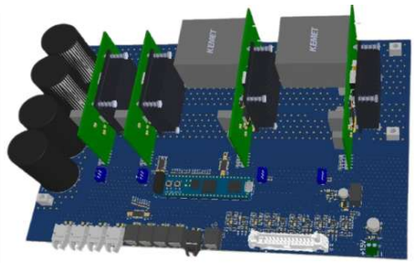
Research in OSIC design

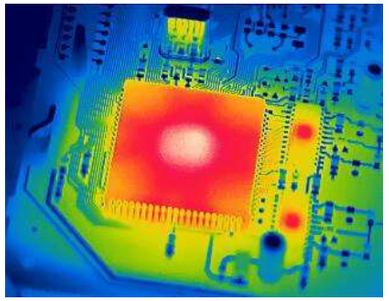
Current work at the Valparaíso ecosystem

Power electronics (PEL) meet integrated circuits (ICs)

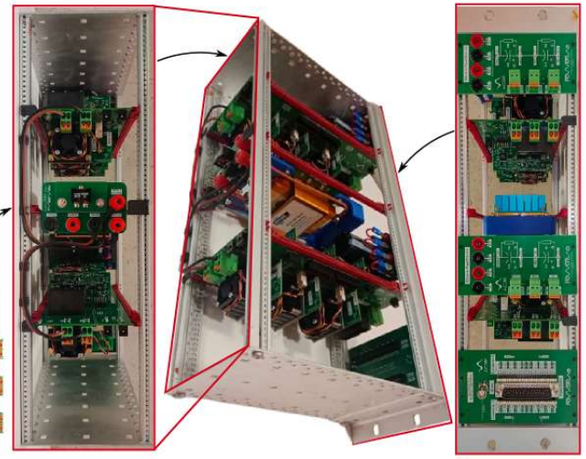




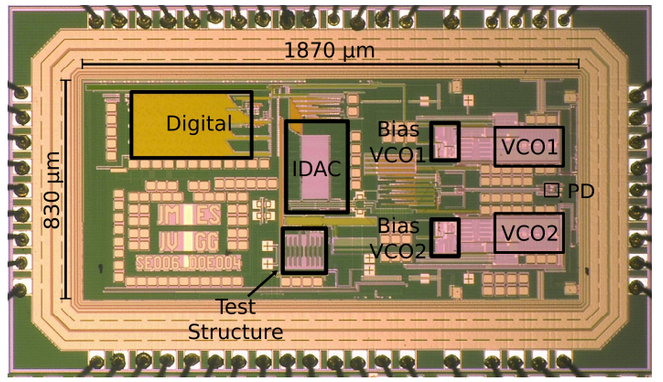




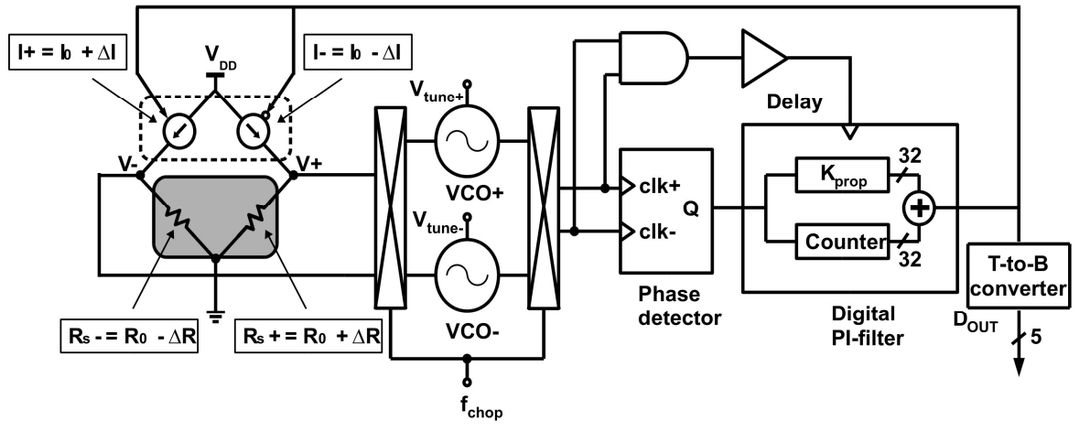




[Rojas et al., IECON2021]







[Marin et al., JSSC2019]

AC3E: 2021-2026 strategy



Synopsys-AC3E training program



EAMTA: Escuela Argentina de Microelectrónica, Tecnología y Aplicaciones



IEEE SSCS Open-Source Chipathon 2022-2026

EUROPRACTICE IC SERVICE

SYNOPSYS®

imec

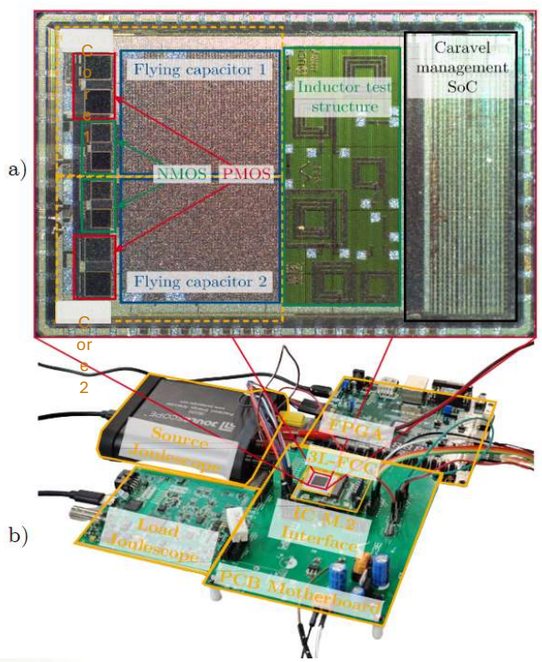
Fraunhofer IISB

ihp innovations for high performance microelectronics

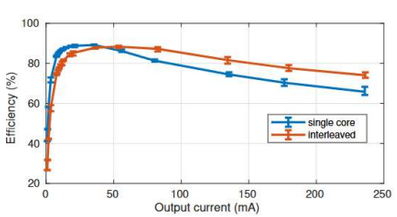
NDAs and research/education agreements

Current research topics at the Valparaíso ecosystem

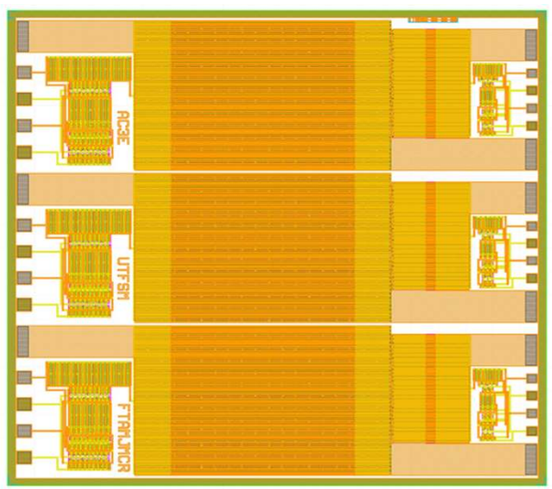
OS design and testing in standard CMOS

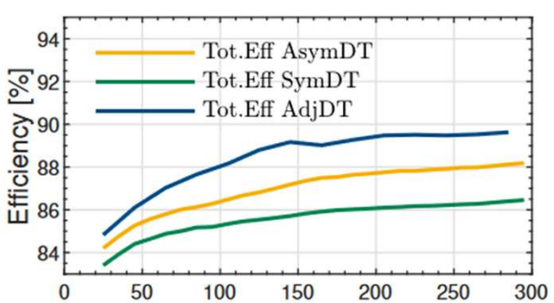




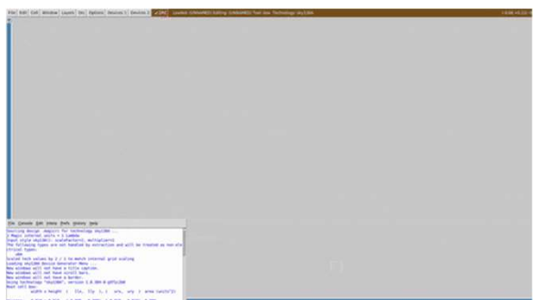


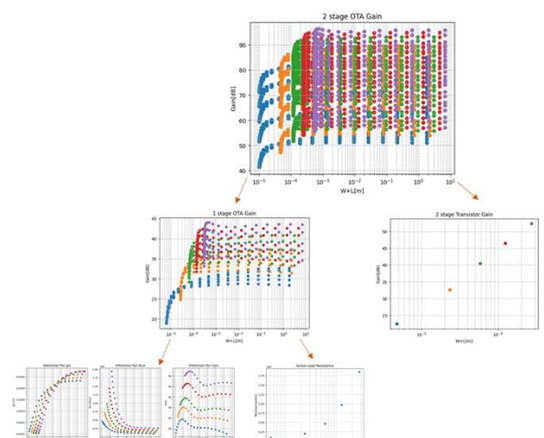
Design of ICs in emerging tech.





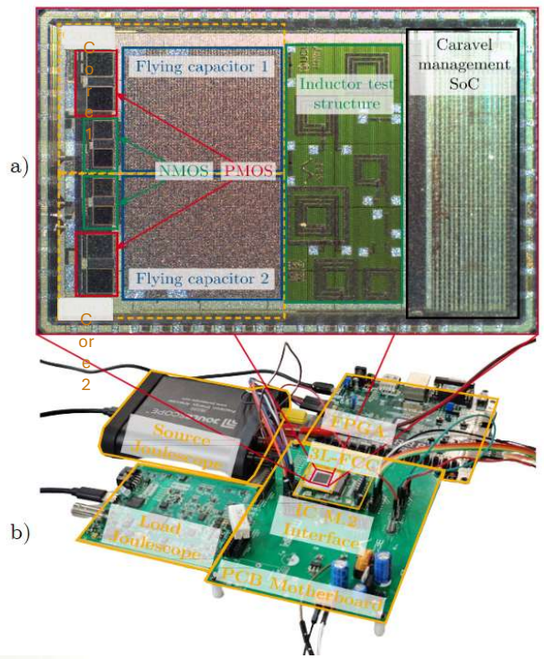
Design automation of PMIC systems



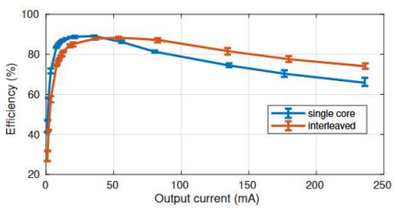


Current research topics at the Valparaíso ecosystem

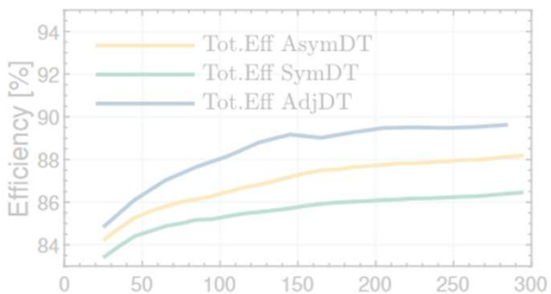
OS design and testing in standard CMOS







Design of ICs in emerging tech.

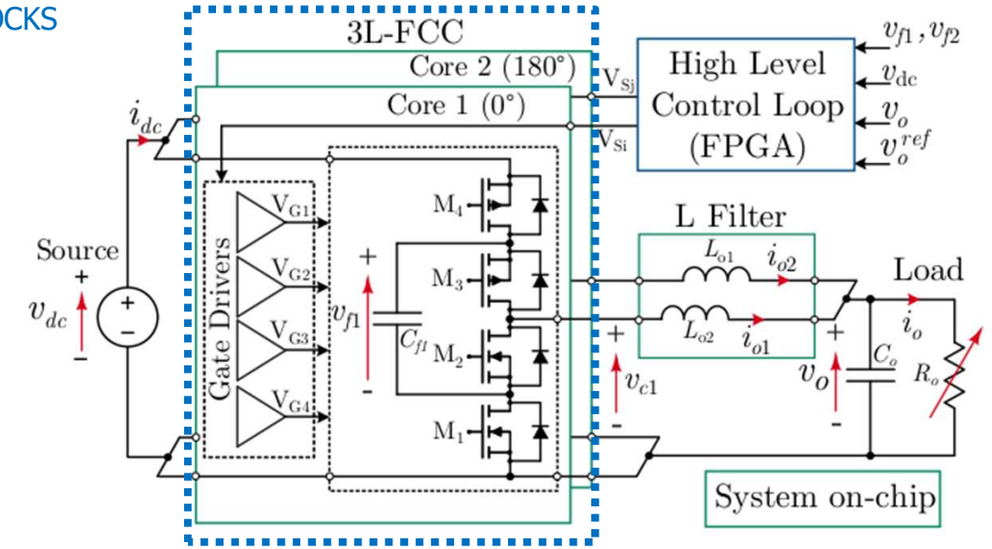


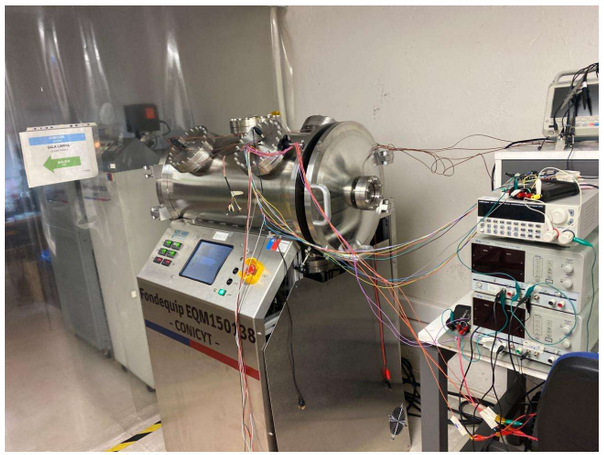
Design automation of PMIC systems

OS design and testing in standard CMOS

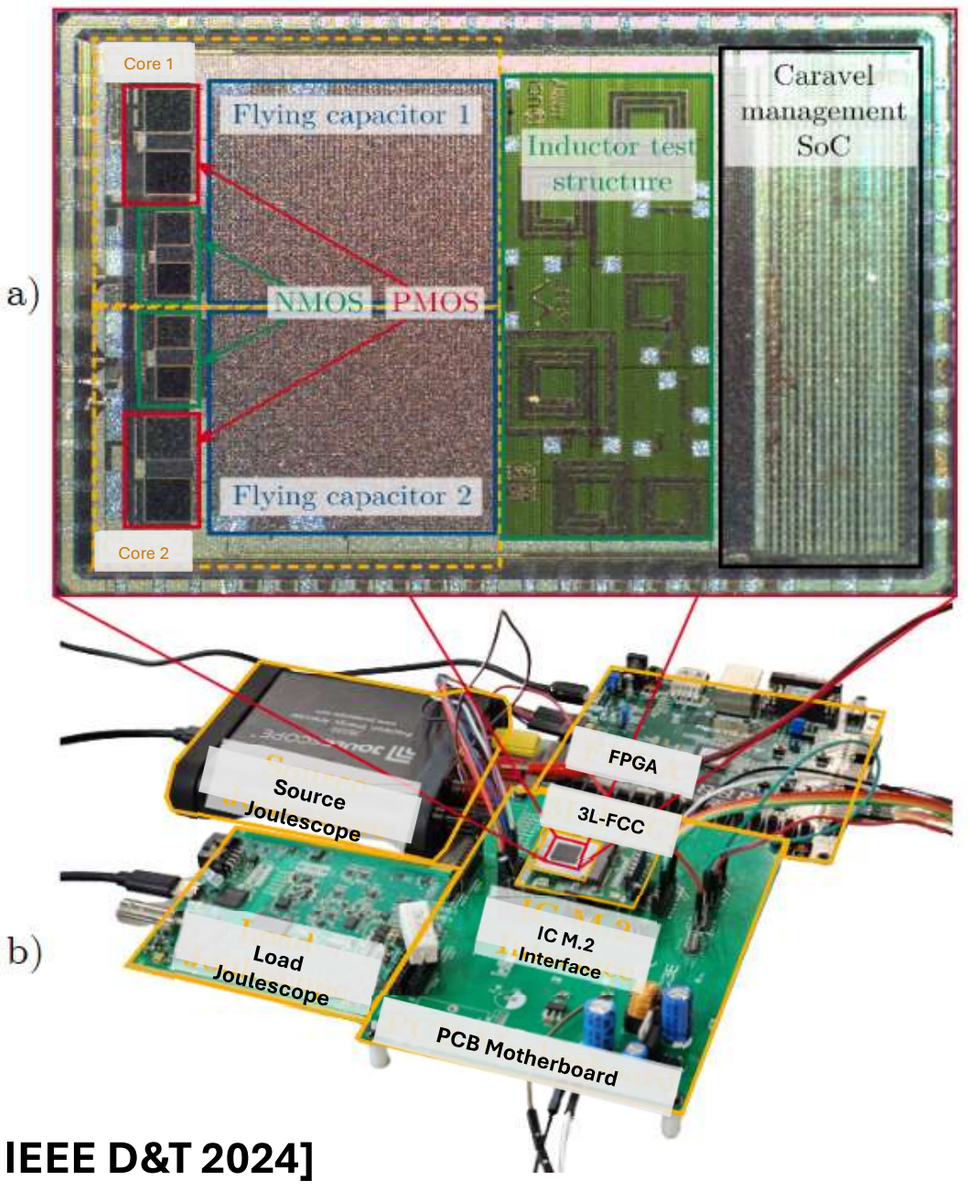
3-level flying capacitor DC-DC converter

ON CHIP BLOCKS





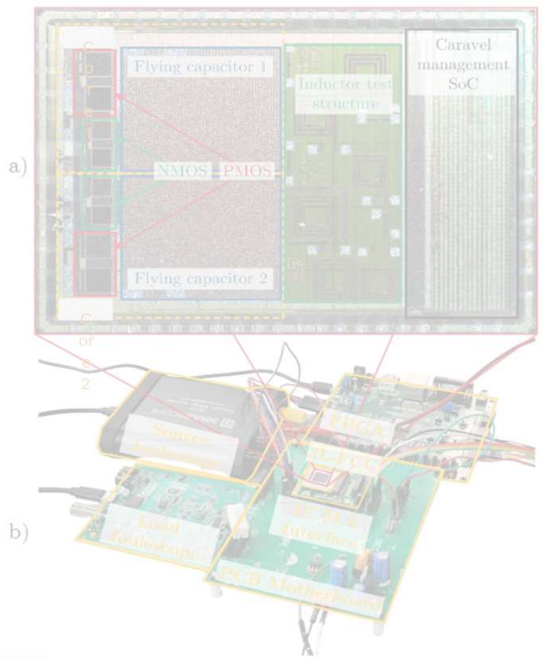




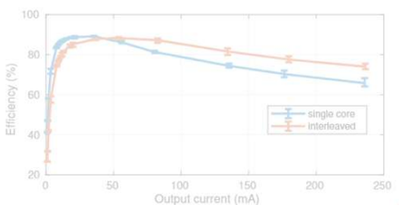
[MARIN, IEEE D&T 2024]

Current research topics at the Valparaíso ecosystem

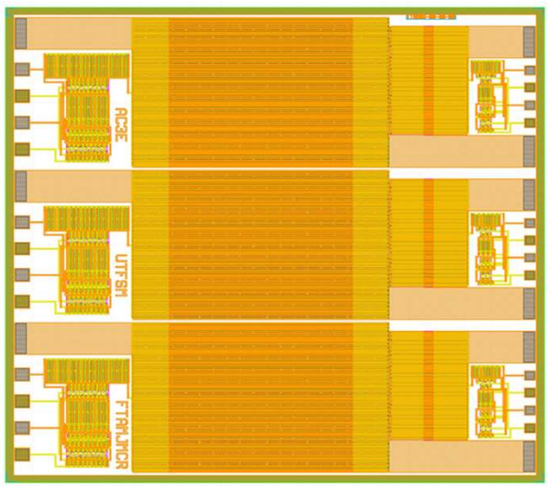
OS design and testing in standard CMOS

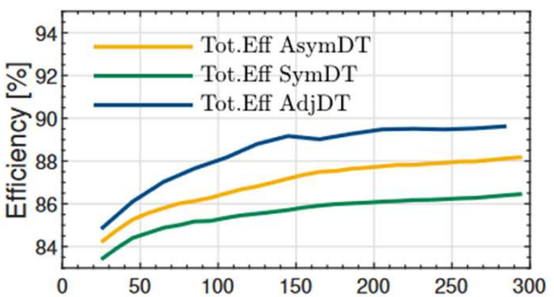






Design of ICs in emerging tech.



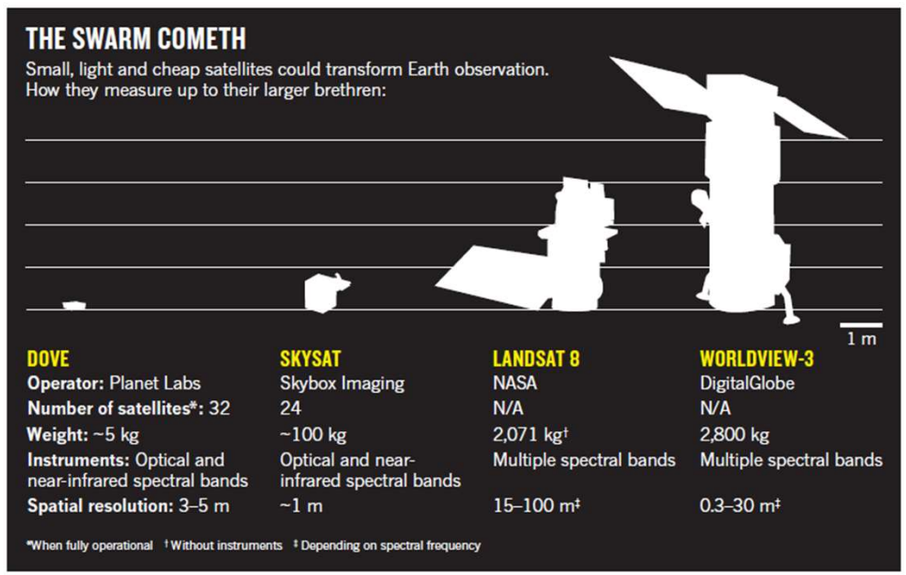


Design automation of PMIC systems

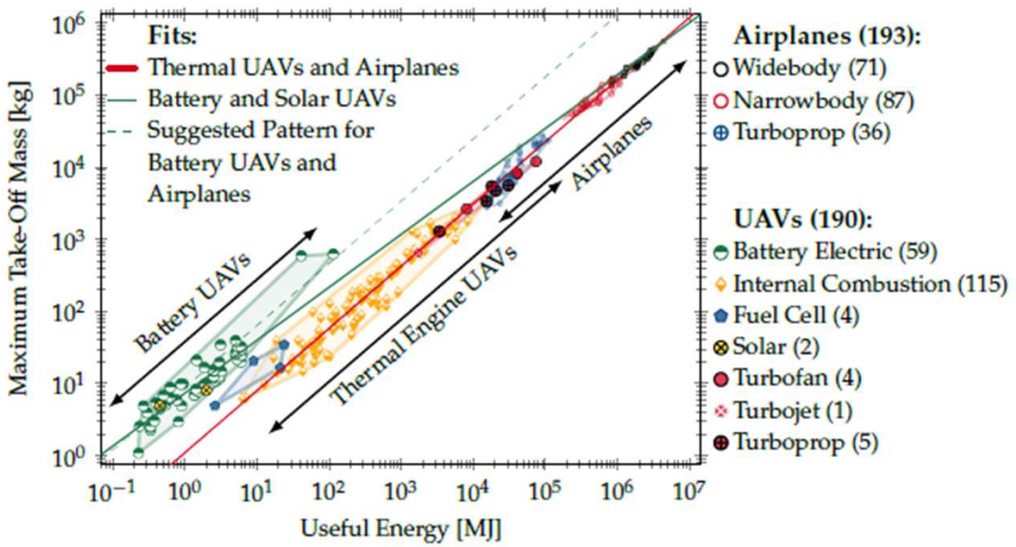
Power and sensor circuits in emerging technologies

PEL + IC challenges

- Conventional power electronics → "99.9% class efficiency"
- Current challenge is **high power density**
- **Electrification** in energy autonomous systems:
  - wide-bandgap materials for **high efficiency** and **demanding environments**
  - on-chip integration of power electronics
  - others, e.g. sensing, thermal management and electromagnetic compatibility



Satellite volume "Moore's law" [Nature, 2016]



UAV mass vs energy plot [MDPI Drones, 2024]

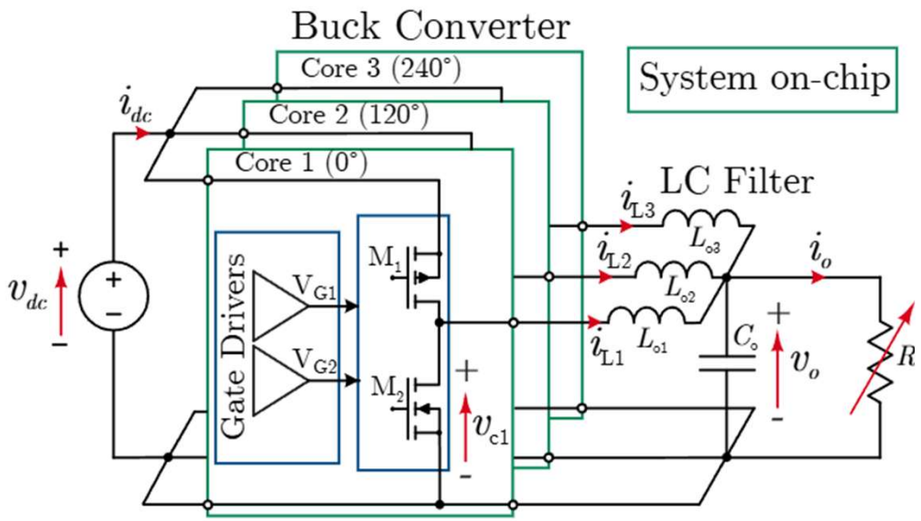
Power and sensor circuits in emerging technologies

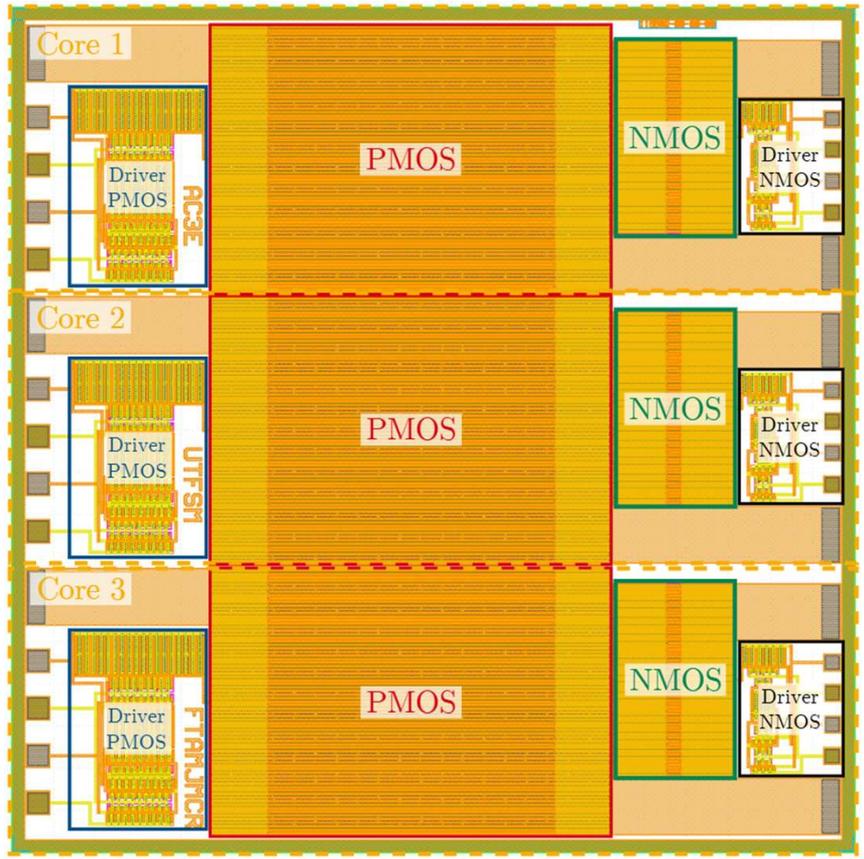
Design with Fraunhofer 4H-SiC technology

3-phase interleaved Buck DC-DC converter

→ Fraunhofer's 2µm 4H-SiC integrated technology

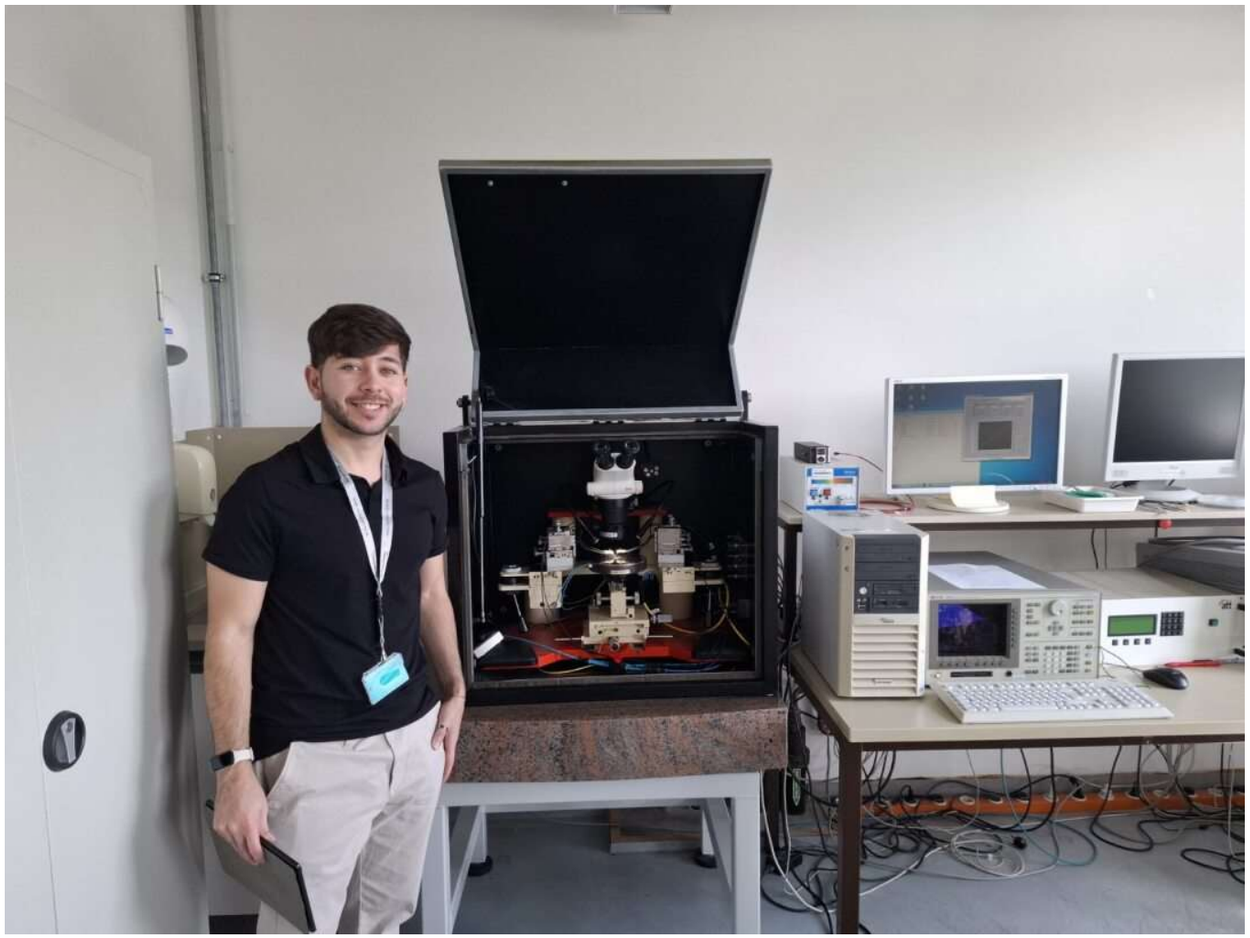
→ Harsh environments: up to 600°C and radiation tolerant operation



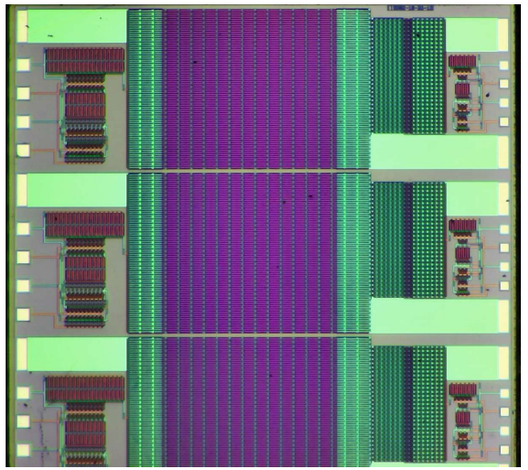


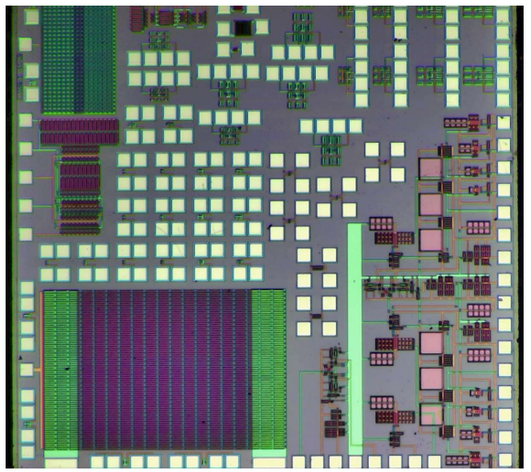
A.Martinez, F. Torres, J. Marin, C. A. Rojas, J. Gak, M. Rommel, A. May, A. H. Wilson-Veas, M. Miguez, C. Rossi, M. Schraml, N. Calarco, "**Analysis and Design of an SiC CMOS Three-Channel DC-DC Synchronous Buck Converter for High-Temperature Applications"**, in Applied Sciences, **2024.**

Power and sensor circuits in emerging technologies



Internship @Fraunhofer: test structure measurements at >500°C

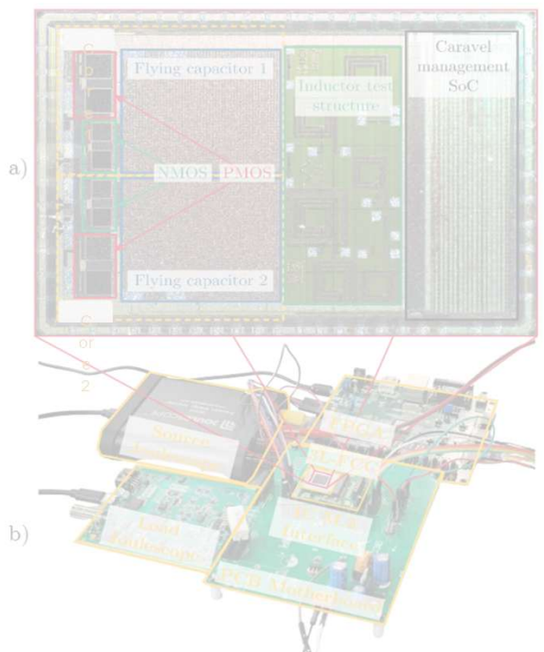




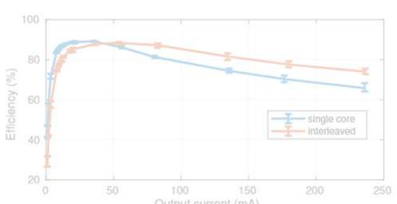
Top: 3-phase SiC buck DC-DC converter
Bottom: Test structures (blocks and sensors)

Current research topics at the Valparaíso ecosystem

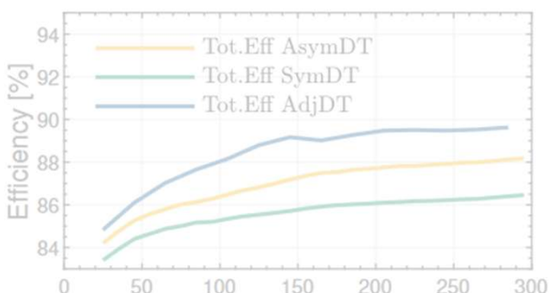
OS design and testing in standard CMOS



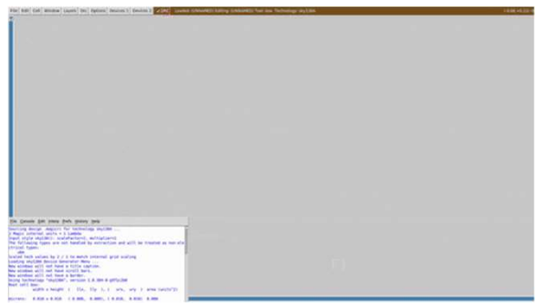


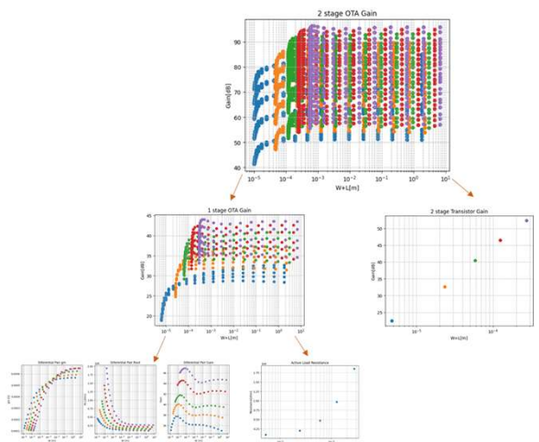


Design of ICs in emerging tech.



Design automation of PMIC systems



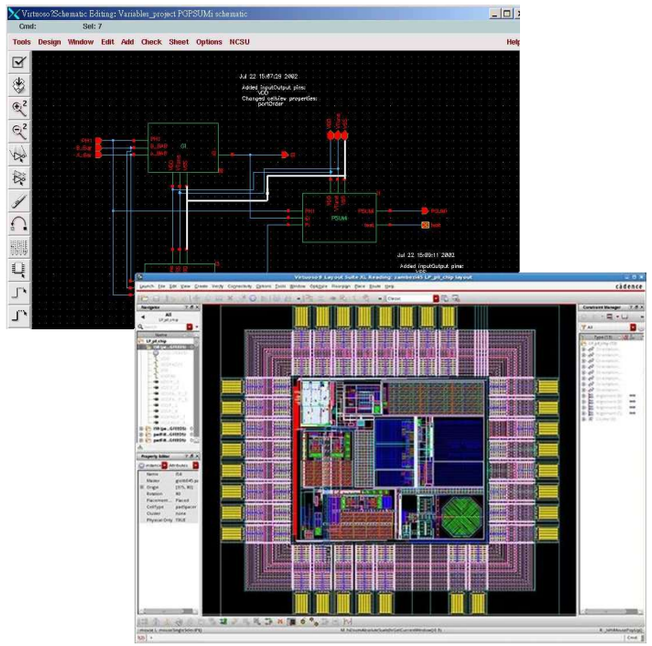


Automation using open source EDA tools and PDKs

Electronic design automation: brief historical perspective



< 1990s → fully manual chip design!



> 1990s → EDA tools

Google + skywater TECHNOLOGY

FOSS 130nm Production PDK
github.com/google/skywater-pdk

IHP-GmbH/**IHP-Open-PDK-docs**

Documentation for IHP 130nm BiCMOS Open Source PDK

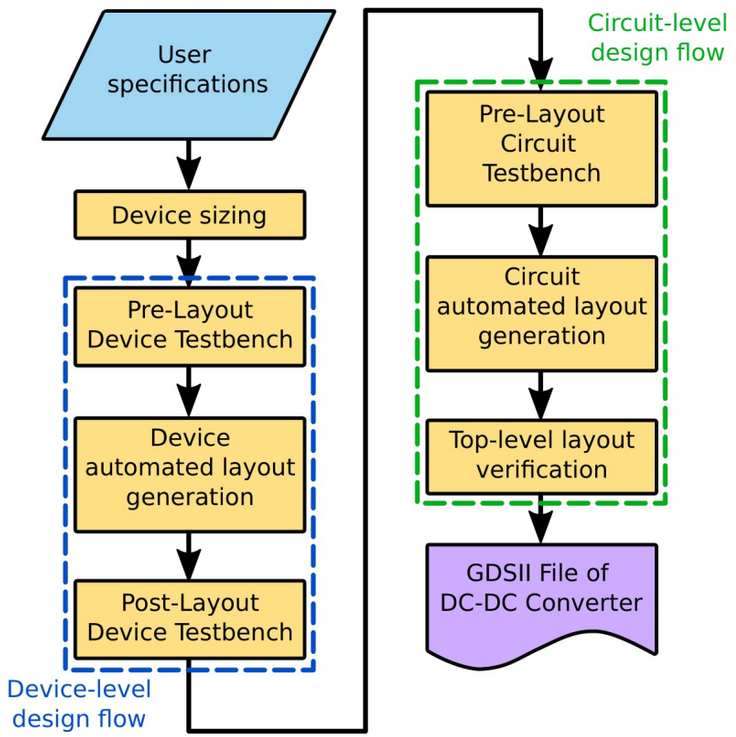
> 2020 → Opensource EDA tools and PDKs

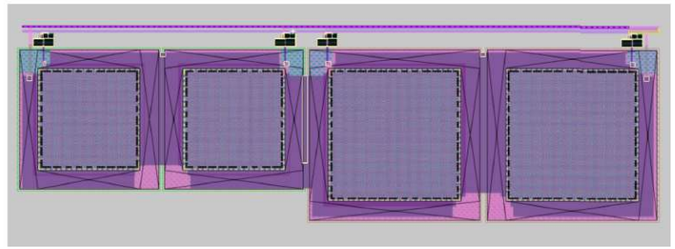
Automation using open source EDA tools and PDKs

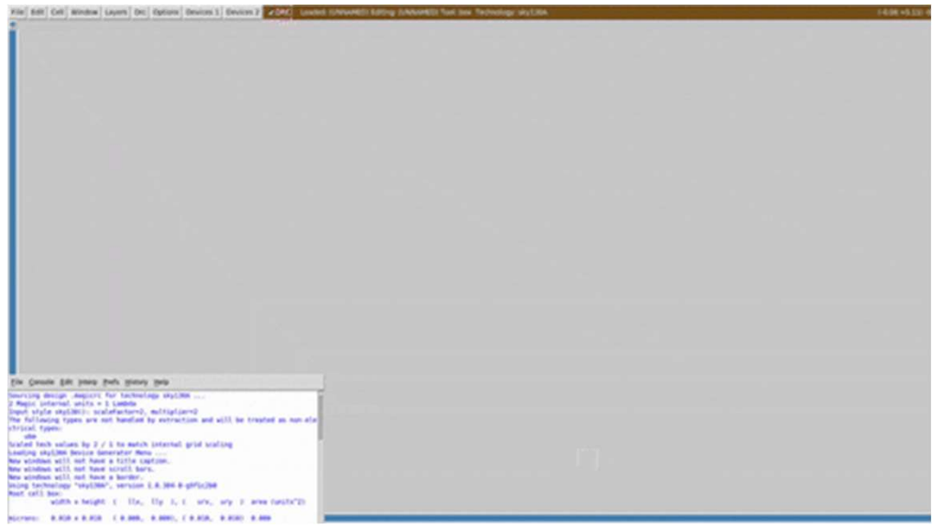
Automated power IC specs-to-GDS flow

- Full Python-based automated flow using opensource toolkit
- From current, voltage, area and frequency specs to manufacturable layout (GDS file)
- Presented at Code-a chip session, VLSI Symp. Kyoto 2023

[OSORIO, CAE 2025]

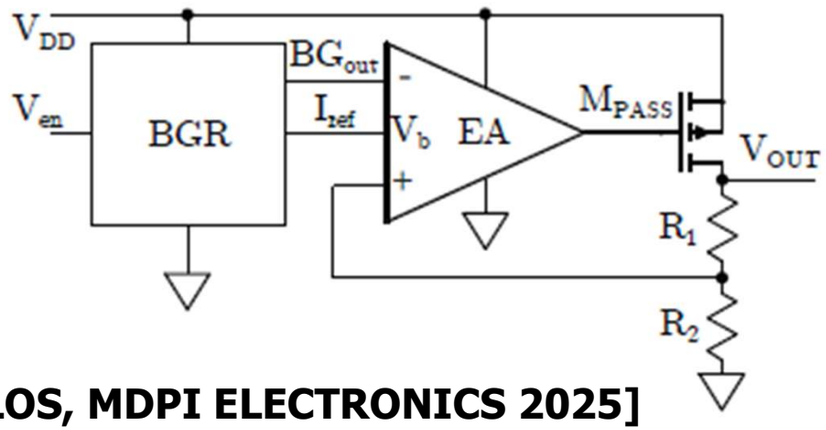




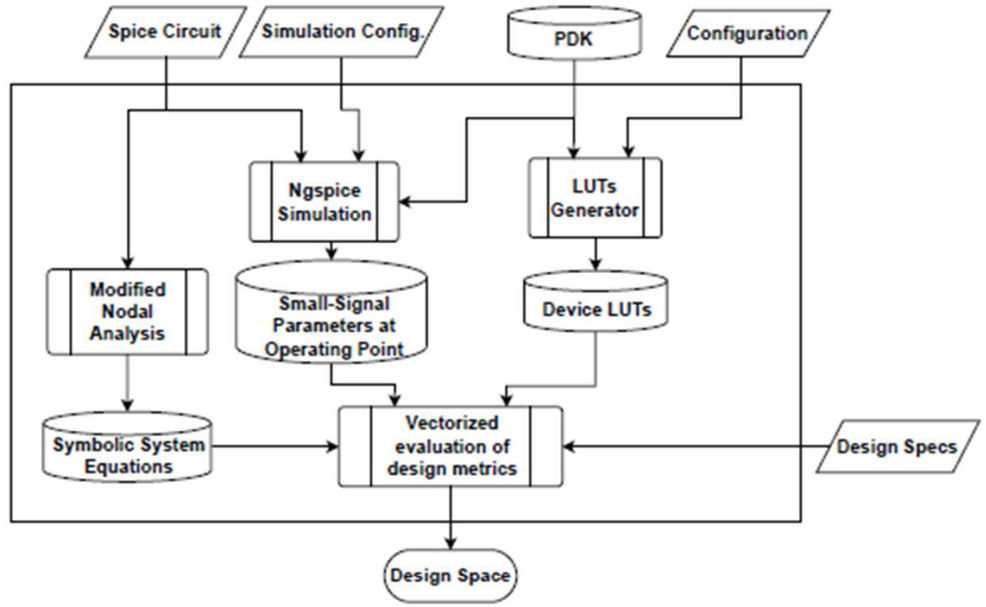


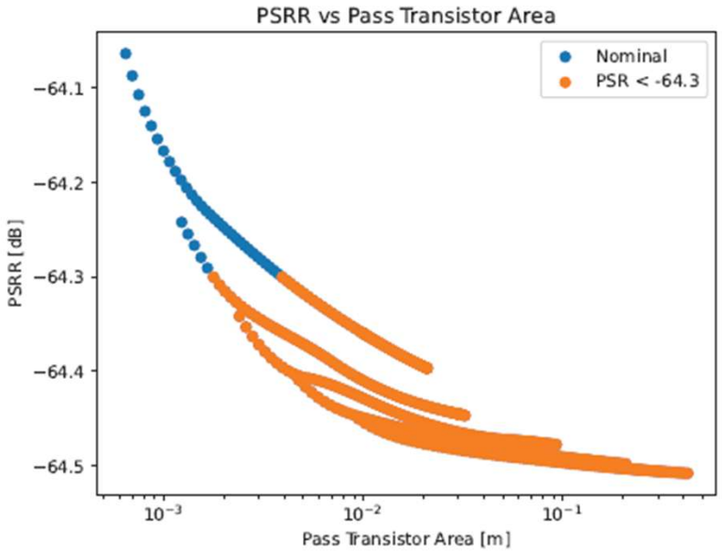
Automation using open source EDA tools and PDKs

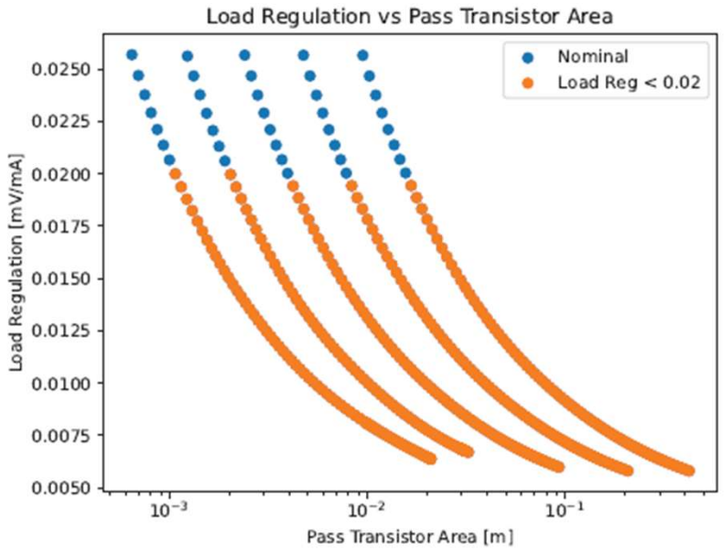
LDO design co-pilot with IHP PDK

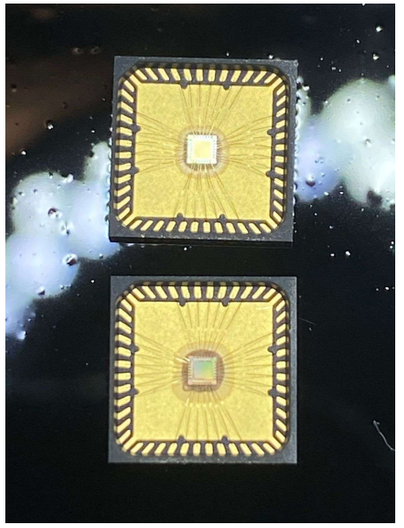


[AREVALOS, MDPI ELECTRONICS 2025]



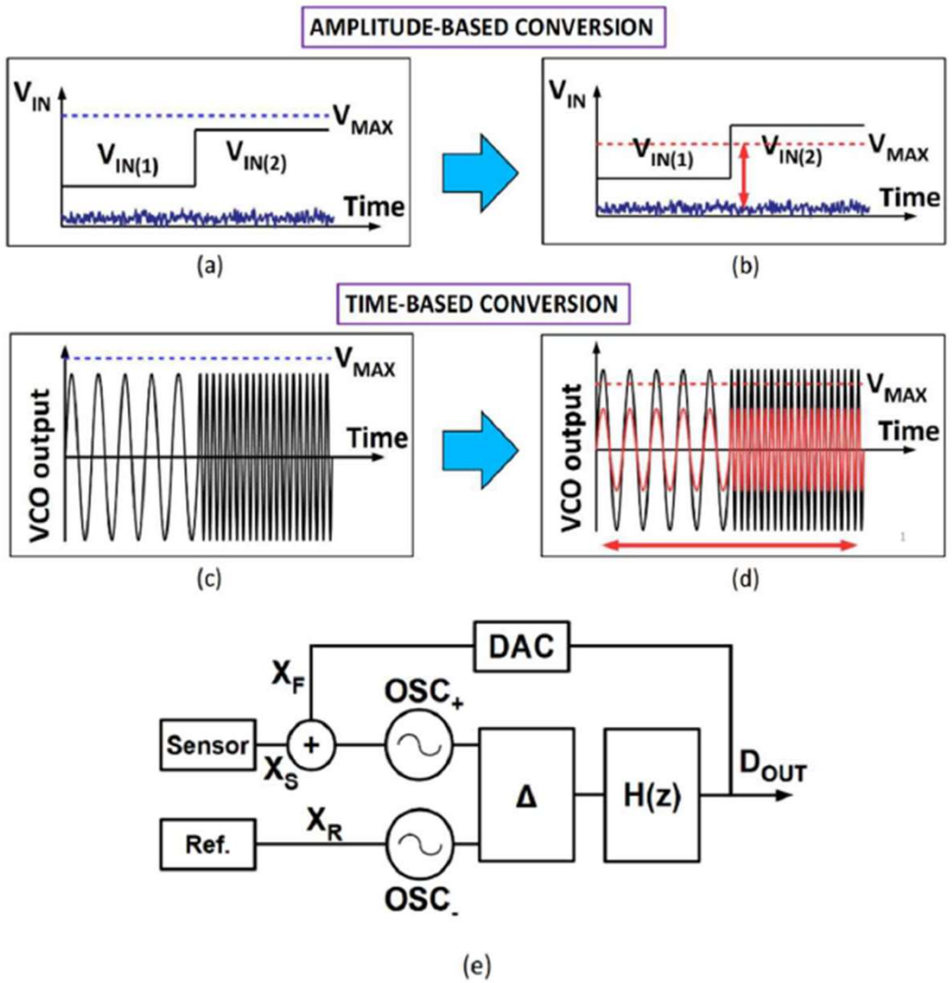






Projects: ANID Iniciación 2024 [Finalized]

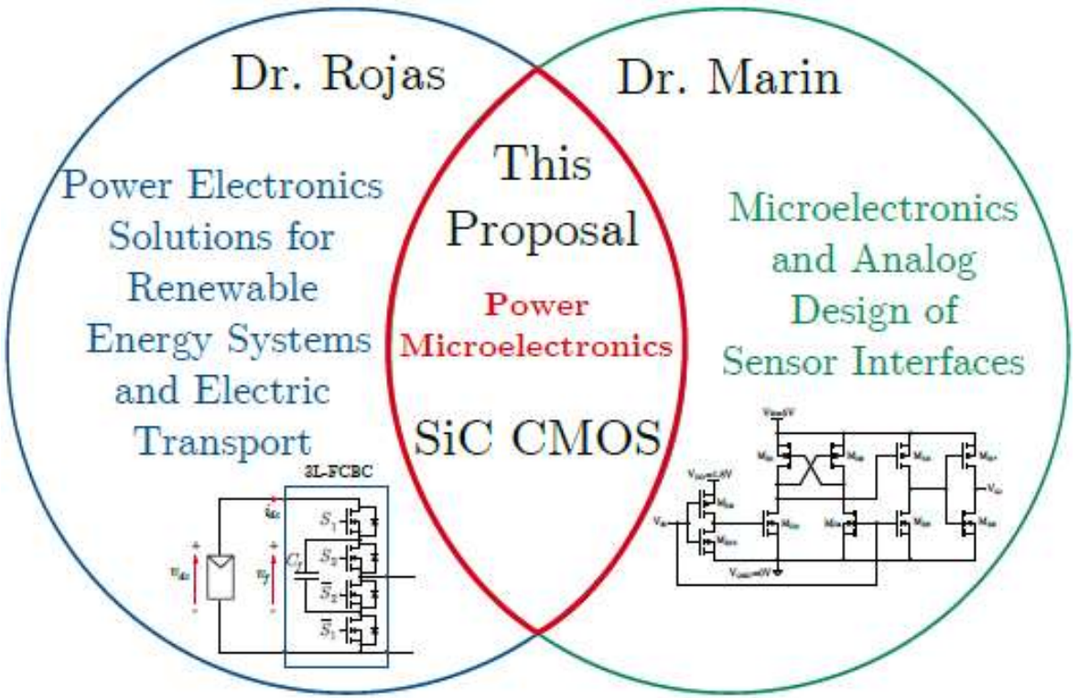
Robust and Area-Efficient Time-Based Sensor Interface Integrated Circuits for Harsh-Environment Applications



- → Two-year project starting on 03/2024, ending on 03/2025
- → Funding: **$70M** ($35M per year)
- → Main objectives:
  - → (SO1): Design of robust time-based sensor interfaces with automated methodologies
  - → (SO2): Develop novel automated design and experimental validation techniques
  - → (SO3): Foster research collaboration and advanced human-capital training in IC design
- → **3 journal papers, 8 conference papers, 4 master thesis, 4 undergrad thesis, collaboration across 5 countries**

Projects: ANID Exploración 2025

Towards a Silicon-Proved Automation Framework of Analog and Mixed-Signal Circuit Designs for Next-Generation Power Integrated Circuit Applications



- → Four year project starting on 08/2025, ending on 07/2029
- → Funding: **$385M** ($96M per year)
- → Main objectives:
  - → (SO1): To design CMOS integrated DC-DC converters
  - → (SO2): To design circuits by using monolithic WBG technologies (SiC or/and GaN)
  - → (SO3): To design sensor interfaces and control circuits for fully integrated DC-DC converters
- → Thesis and undergraduate research topics available

Projects: IHP-AC3E Collaboration Agreement

End-to-end, proven hands-on experience

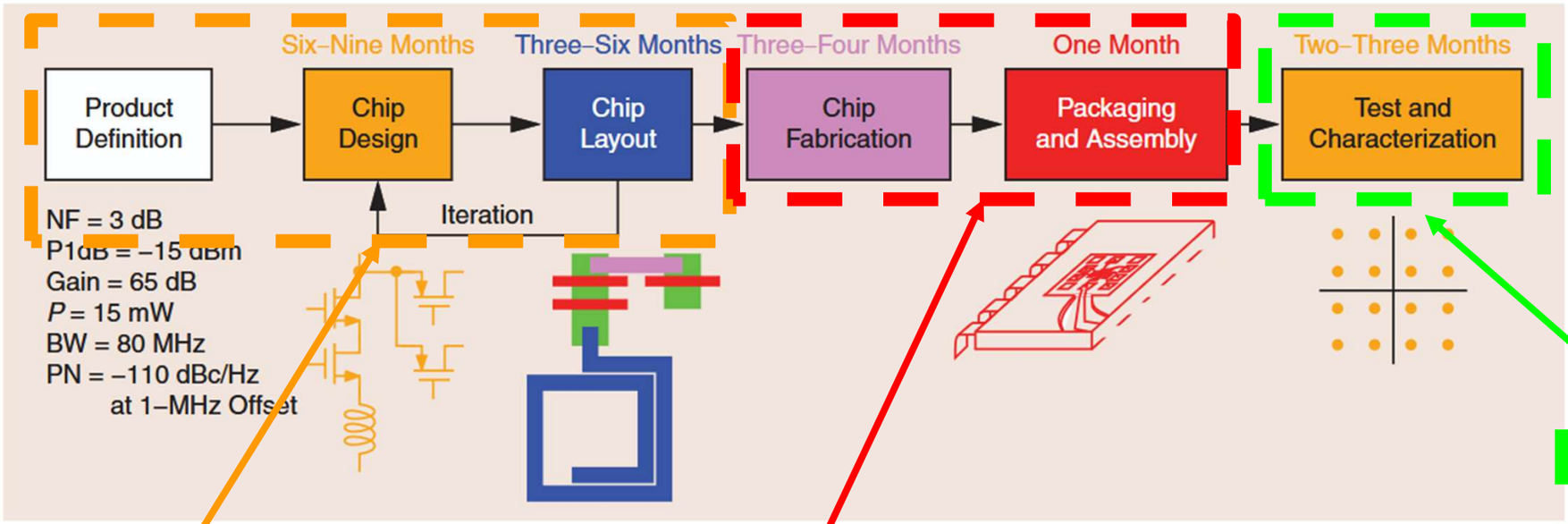


FIGURE 1: A typical chip development time frame.

[RAZAVI, SSCM2024]

AC3E

Joint characterization

→ Three year project, Funding: **$100M** ($33M per year)

Projects: IHP-AC3E Collaboration Agreement

Success stories: Daniel Arévalos' (USM) internship at IHP (April-June '24)



EX UMBRA IN SOLEM

UNIVERSIDAD TECNICA
FEDERICO SANTA MARIA

Tesis de Magíster

Automatización del diseño de reguladores de bajo drop-out en sistemas DC-DC integrados: una metodología estructurada y sistemática independiente de la tecnología de fabricación

Leveraging Lookup Tables for Efficient LDO Design Exploration using Open-Source CAD Tools and IHP-Open130-G2 PDK

Daniel Arevalos[1], Jorge Marín[1,2], Krzysztof Herman[3], Christian A. Rojas[1,2]
[1]Department of Electronic Engineering, Universidad Técnica Federico Santa María, Valparaiso, Chile,
[2]Advanced Center of Electrical and Electronics Engineering, Universidad Técnica Federico Santa María,
daniel.arevalos@sansano.usm.cl, jorge.marinn@usm.cl
[3]IHP – Leibniz Institute for High Performance Microelectronics, Im Technologiepark 25, 15236 Frankfurt (Oder), Germany
herman@ihp-microelectronics.com

Open Access Article

A Topology-Independent and Scalable Methodology for Automated LDO Design Using Open PDKs

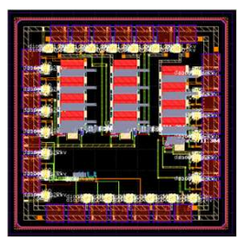
by Daniel Arévalos [1,*], Jorge Marin [2,3,*], Krzysztof Herman [4], Jorge Gomez [5], Stefan Wallentowitz [1] and Christian A. Rojas [2,3]

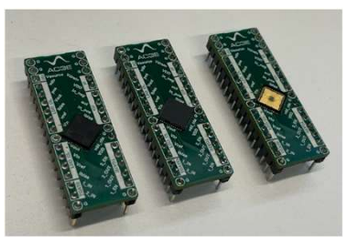
EX UMBRA IN SOLEM

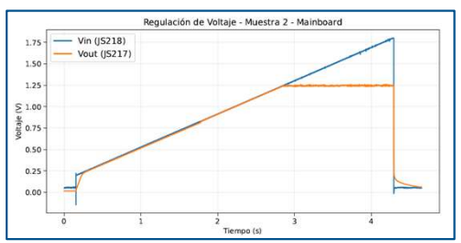
UNIVERSIDAD TECNICA
FEDERICO SANTA MARIA

Memoria de Título

Desarrollo de un Entorno Experimental para la Validación de un Circuito Integrado CMOS Customizado



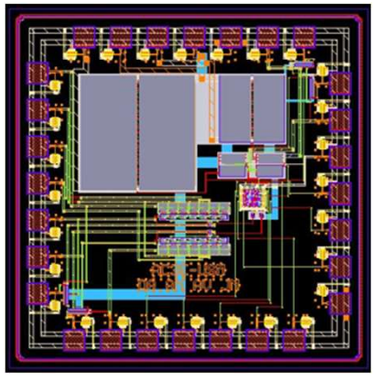




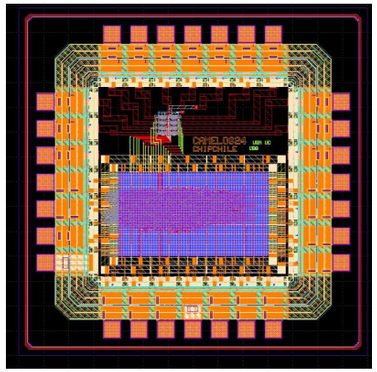
Projects: IHP-AC3E Collaboration Agreement

Success stories: IHP Workshop at USM, 30 Sep - 2 Oct 2024





Time-based DC-DC



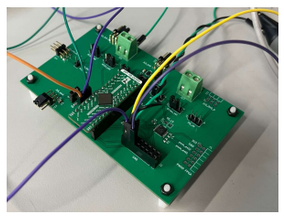
Time-based AI accelerator

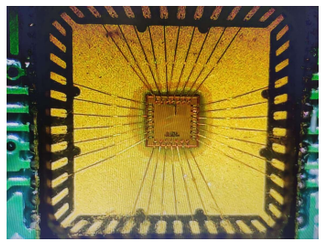
Scope:

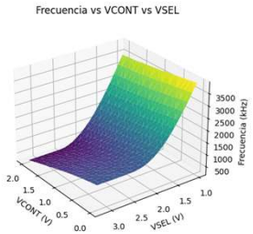
- digital track
- analog mixed signal track
- designs were submitted in TO_Nov2024
- currently under measurement

Tapeout-Oriented Strategies for Teaching Analog/Mixed-Signal IC Design and Implementation in Electrical Engineering Education — 2025
J Marin, K Herman, D Arévalos, M Andraud, CA Rojas
2025 38th SBC/SBMicro/IEEE Symposium on Integrated Circuits and Systems …

Systematic Design of a PVT-Robust CMOS Time-Based-Controlled DC-DC Converter Using Open-Source Tools — 2025
J Marin, V Osorio, A Martinez, D Arévalos, K Herman, JPM Brito, CA Rojas
2025 IEEE Nordic Circuits and Systems Conference (NorCAS), 1-6





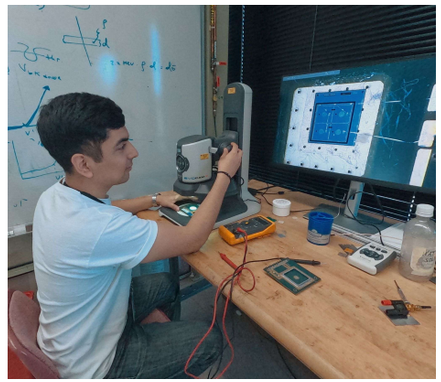


Other success stories: graduates in microelectronics



FIGURE 2: A snapshot of the Code-a-Chip winners with Dr. John R. Long, SSCS president (right).

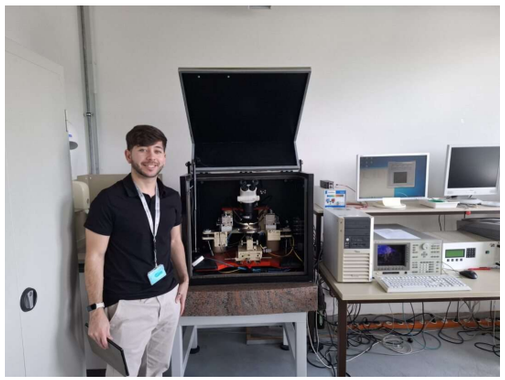
Vicente Osorio
PhD Candidate @TU\e, Netherlands



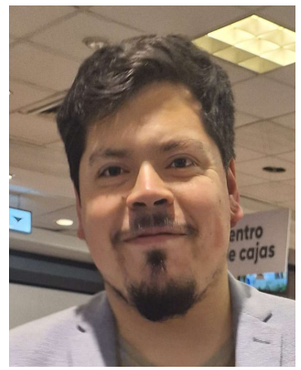
Nelson Salvador
Internship @Fermilab



Max Vega
Scholarship @UNISINOS, Brazil



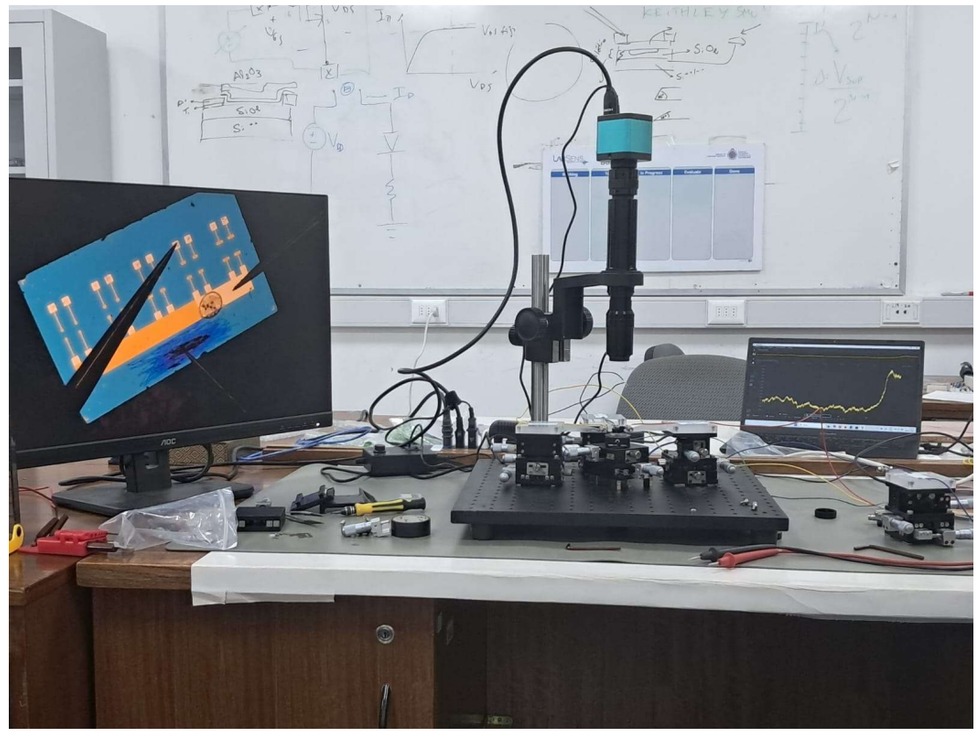
Andrés Martínez
Project Engineer @AC3E



Daniel Arévalos
PhD Candidate @HM, Germany

Other ongoing research at the Valparaíso ecosystem





Probe station capabilities at EIE-PUCV

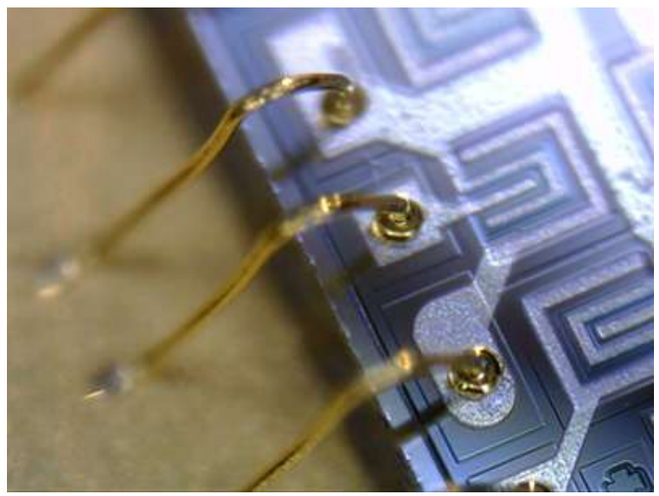
What's next?

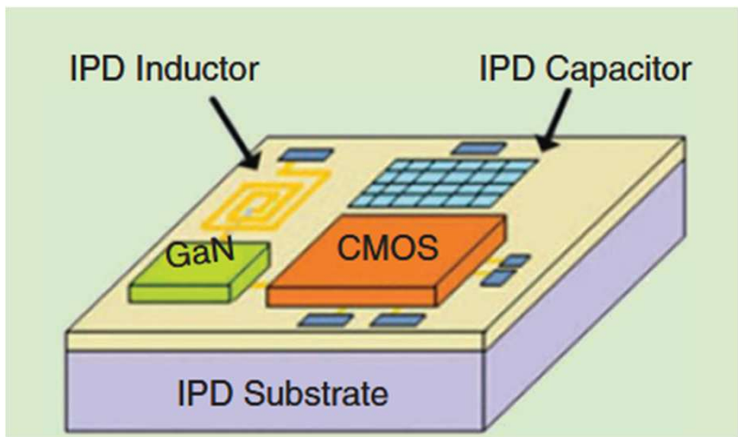
→ Our aim: become a **reference center** for **highly-integrated systems** at **regional and global level**

- Training and **advanced human capital**
- OS and commercial (hybrid) **automation-focused design methodologies**
- Our core capability: silicon validated designs from concept to lab
  - Advanced assembly+packaging → **heterogeneous integration**







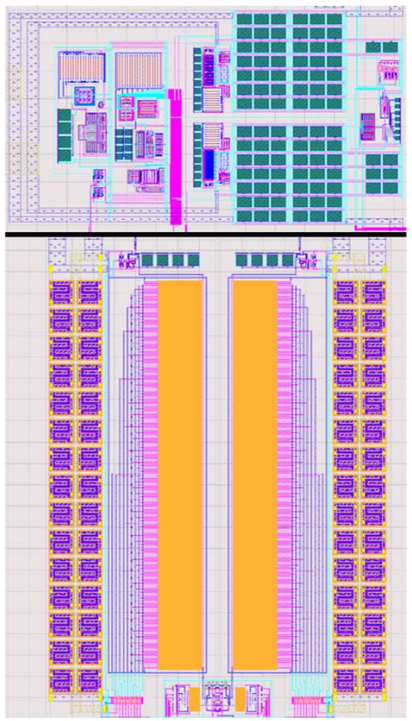


Current opportunities for emerging teams

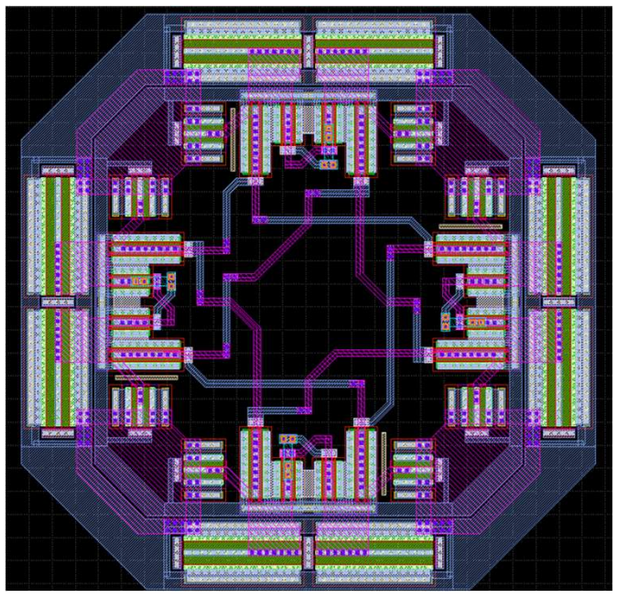
Other team's R&D in OSIC design

OPENSOURCE DESIGNS/IP

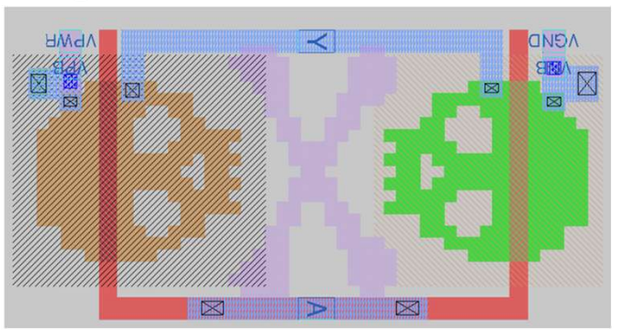
SAR ADC



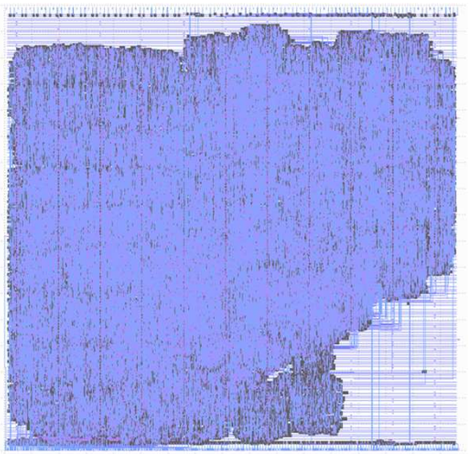
Satellite transceiver



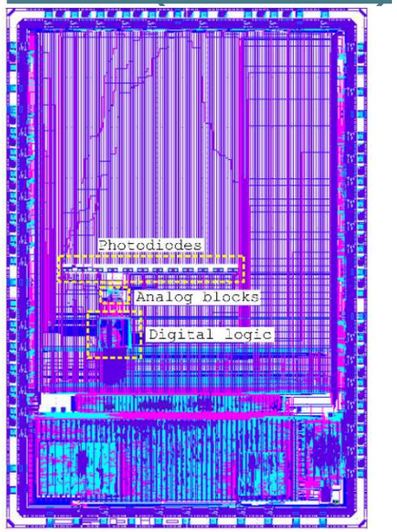
Logic inverter



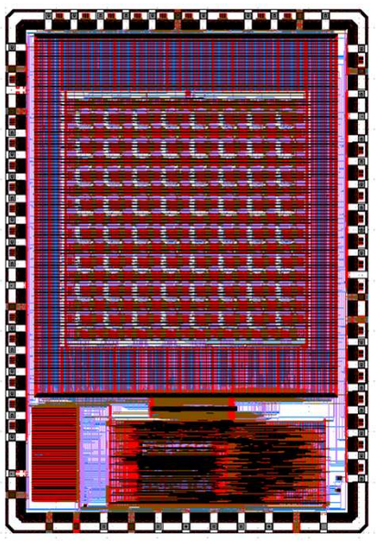
Sudoku Accelerator



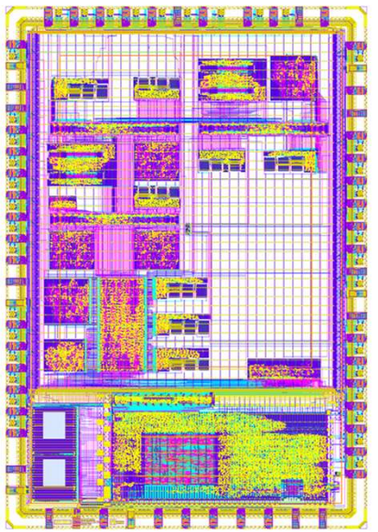
MixPix (**U. Biobio**)



OpenFPGA



Time to Data Converter



Opensource designs → re-use and get inspired

- Analog IP components and specifications in Sky130A for Efabless next-generation Caravel (Chipalooza): https://opencircuitdesign.com/chipalooza/leaderboard-2.html

Rank	Name	Circuit Type	Repository
1	Shivaram Mysore (Vyges)	PLL (self-biased ring-oscillator)	View repository
2	rcyaon	Level shifter (1.2V<->3.3V, 4-channel)	View repository
3	Team MAKers	ADC (10-bit SAR, differential)	View repository
4	prajapati	SERDES RX (programmable CTLE + reference-less bang-bang CDR)	View repository
5	brahim	Voltage supervisor (POR + BOR)	View repository
6	kvitschal	CMOS voltage reference	View repository
7	Shivaram Mysore (Vyges)	LDO (capless, digitally-trimmed)	View repository
8	malik	Programmable-Gain Instrumentation Amplifier	View repository

- IHP opensource designs (per tapeout), e.g.: https://github.com/IHP-GmbH/TO_Nov2024

Opensource chip fabrication initiatives

- Chipalooza: https://opencircuitdesign.com/chipalooza
- IEEE SSCS Chipathon: Build → Test → Publish educational program
  → https://sscs.ieee.org/technical-committees/tc-ose/sscs-pico-design-contest/
- IEEE UNIC-CASS: Universalization of IC design
  → https://ieee-cas.org/universalization-ic-design-cass-unic-cass
- IEEE Open Silicon Initiative: sponsors Tiny Tapeout blocks
  → https://technical-community-spotlight.ieee.org/democratizing-microelectronics-ieee-division-i-open-silicon-initiative/

About our 2-day workshop

Introduction to Analog Integrated Circuit Design using Open-Source EDA Tools and Open PDKs

Goal

Provide practical, reproducible training in analog IC design using open-source EDA tools and open PDKs.

Target Audience

- Undergraduate Students
- Graduate Students
- Faculty Members
- Researchers

Workshop Activities Overview

1. **Schematic Capture** — Design and configure an analog circuit (e.g., amplifier, LDO, or building block).
2. **Transistor-Level Simulation** — Analyze circuit performance (DC, AC, transient) and iterate the design.
3. **Physical Layout** — Create the layout and apply design rules (DRC).
4. **Verification** — DRC, LVS, PEX. Verify correctness (DRC/LVS) and assess post-layout performance (PEX).
5. **Automation-Oriented Design** — Introduction to scripting and design automation for reproducible flows.

2-Day Agenda

Day 1 – Design and Simulation

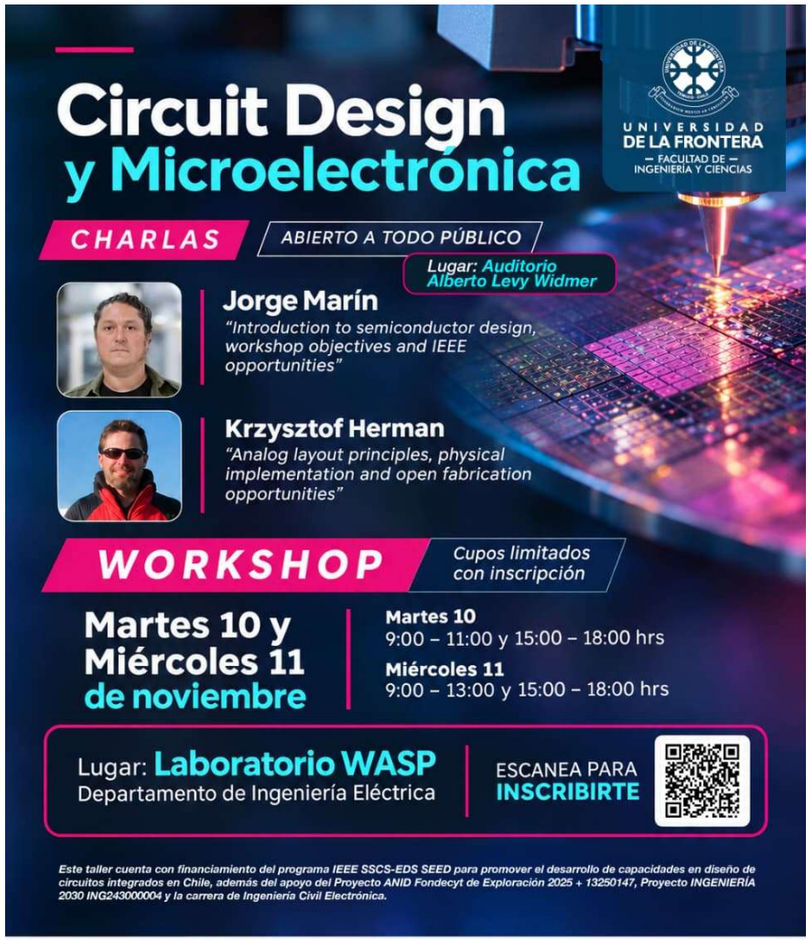
- Introduction to open-source EDA tools and open PDKs
- Analog circuit design and schematic capture
- Transistor-level simulation and design iteration
- Hands-on laboratory session

Day 2 – Layout, Verification and Beyond

- Physical layout design
- Design verification (DRC/LVS) and post-layout analysis
- Introduction to design automation
- IEEE opportunities: Chipathon, Code-a-Chip, Chip-practice and UNIC-CASS
- Wrap-up and next steps

Expected Outcomes

- Hands-on experience in a complete analog IC design flow.
- Skills and resources to continue IC design and participate in IEEE initiatives.
- Reusable educational materials for future university courses.



Thank you!
Questions?

EXAMPLES FROM OTHER TEAMS

- D&T edition
- 3-bit flash SKY130